

NEURAL REGRESSION AND RANDOMIZED OPTIMIZATION FOR IMPULSE CONTROL PROBLEMS

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Abstract. We develop a regression-based Monte Carlo framework for stochastic impulse control problems with controls taking values in a continuous compact set. The approach combines dynamic programming on a discrete intervention grid with neural-network approximations of continuation values and randomized optimization procedures for intervention operators. We first establish a time-discretization result for δ -separated strategies, proving an approximation error of order $O(|\pi|^{1/2})$. On the discrete grid, we derive a dynamic programming representation of the solution in terms of intervention regions and recursively defined stopping rules, together with a verification lemma characterizing optimal strategies. Building on this representation, we introduce a backward numerical algorithm in the spirit of regression Monte Carlo methods. Finally, relying on recent approximation results for piecewise affine neural networks, we establish convergence estimates for the randomized approximation of intervention operators. Numerical experiments on classical dividend and harvesting problems illustrate the robustness of the proposed methodology in both low and moderately high dimensions.

Key words. impulse control, stochastic control, regression Monte Carlo, neural networks, randomized optimization, dynamic programming

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1. Introduction. Stochastic impulse control provides a versatile framework to model systems whose dynamics combine a continuous stochastic evolution with discrete and instantaneous interventions. Such problems arise in mathematical finance, economics, operations research, and natural-resource management. Classical applications include optimal dividend distribution [41, 16, 22], algorithmic trading [13], optimal execution [29], portfolio or inventory adjustments under transaction costs [32, 15, 40], irreversible investment [24], control of exchange rates [28, 17], and harvesting problems [4, 41, 5, 37].

From a theoretical perspective, impulse control has been extensively studied over the past decades. Classical contributions established the foundations of the theory through quasi-variational inequalities, verification arguments, and structural characterizations of optimal strategies; see in particular [11, 21, 41, 43]. Subsequent developments expanded the analytic framework in several directions, notably through viscosity-solution techniques [8, 12, 6] and connections with singular stochastic control [5]. Overall, impulse control benefits from a broad and robust theoretical foundation. In contrast, the numerical analysis of impulse control remains comparatively underdeveloped. As emphasized by Ludkovski in [39], the available literature is sparse when compared with that on optimal stopping or continuous control. QVI-based methods face several well-known difficulties, including the non-locality of the intervention operator, the intricate geometry of free boundaries, and regularity assumptions that are often difficult to verify. Nevertheless, in more structured settings such as optimal switching problems, where the admissible control space is finite, several numerical methodologies have been successfully developed; see for instance [3, 9, 26]. More gen-

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erally, alternative numerical approaches have also been investigated, including stochastic mesh techniques [14], quantization methods [42], and neural-network-based schemes [31, 33, 10].

In the present work, we remain in the general framework where impulse controls take values in a compact subset of $\mathbb{R}^q$. From a numerical perspective, our approach relies on regression methods exploiting the dynamic programming structure of the discretized problem. Among regression-based approaches, conditional expectation methods originate from the pioneering work of Tsitsiklis and Van Roy [47], while the Longstaff–Schwartz algorithm [38] remains one of the most influential Monte Carlo methodologies, with its probabilistic interpretation in terms of stopping times later clarified [19].

As in classical optimal stopping problems, our numerical methods still relies on the approximation of conditional expectations, which constitutes a central difficulty of dynamic programming methods [33]. The key additional challenge in impulse control comes from the intervention mechanism itself: at each potential intervention time, one must not only decide whether to intervene, but also determine the optimal impulse within a continuous control space. In this dual context, neural networks emerge as natural candidates. Their universal approximation capabilities [20, 27, 30] make them effective tools for learning complex continuation values. Moreover, the recent results of Abbas-Turki and Li [34] provide a control of the error on a global maximum when intervention gains are approximated by random search on piecewise-linear neural networks, ensuring a reliable treatment of the local optimization step. These architectures also align naturally with modern GPU and TPU hardware [45, 18], making it possible to exploit large-scale parallel computation for both components of the problem. More sophisticated optimization methods do exist, including actor–critic and policy-gradient algorithms such as DDPG [36] and PPO [44], as well as derivative-free strategies like CMA-ES [25]. While elegant and powerful in their respective domains, these approaches introduce an additional algorithmic layer that is unnecessary here. The randomization-based maximization adopted in this paper provides a particularly simple and effective alternative: it integrates seamlessly with the dynamic programming structure, avoids algorithmic overhead, and offers a robust and easily implementable way to approximate optimal impulses.

This unified framework, combining neural approximation for conditional expectations and uniform search for impulse selection, thus provides a natural foundation for developing flexible and efficient numerical schemes for impulse control.

Our contributions are fourfold. First, we introduce a time-discretization framework for continuous-time impulse control problems. To rule out impulse clustering phenomena, we consider δ -separated strategies and show that they can be approximated by grid-valued controls with an error of order $O(|\pi|^{1/2})$. Second, once the problem is restricted to a discrete intervention grid, we derive a dynamic programming representation of the solution in terms of stopping times and intervention regions. This construction leads to an explicit characterization of optimal strategies through recursively defined stopping rules, together with a verification lemma identifying the associated value functions. We also extend this framework to the case where the controller is constrained to perform only finitely many interventions. Third, we develop a backward numerical scheme based on Monte Carlo regression methods in the spirit of [47, 38]. The continuation values are approximated recursively by neural networks, while the intervention step requires solving a local optimization problem over the impulse space. The resulting algorithm naturally extends to the finite-intervention

setting through an additional state variable tracking the remaining intervention budget. Fourth, relying on recent results of [34], we establish convergence estimates for the randomized approximation of intervention operators when continuation values are represented by piecewise affine neural networks. We finally illustrate the performance of the resulting scheme on several classical impulse control problems, including dividend distribution and harvesting models.

The paper is organized as follows. Section 2 introduces the continuous-time impulse control framework and proves a reduction to finite-intervention strategies. Section 3 establishes the time-discretization result for δ -separated controls. Section 4 develops the discrete-time dynamic programming formulation of the problem, introduces the associated intervention regions and stopping rules, proves a verification result characterizing the value function and optimal strategy on the discrete grid, and extends the construction to the finite-intervention setting. Section 5 then presents the associated regression-based numerical algorithm together with its finite-intervention extension. Section B studies the convergence of the randomized maximization procedure used to approximate intervention operators. Finally, Section 6 presents numerical experiments on the well known optimal dividend problem [41, 16, 22] in the one-dimensional setting and in higher-dimensional regimes.

2. Notation and finite intervention. Throughout this paper, we are given a finite terminal time $T > 0$ and a probability space $(\Omega, \mathcal{G}, \mathbb{P})$ endowed with a filtration $(\mathcal{F}_t)_{t \geq 0}$ and a d -dimensional standard $\mathcal{F}$ -Brownian motion $W = (W_t)_{t \geq 0}$ for some integer $d \geq 1$. We also fix an integer $m \geq 1$ and denote by $\mathbf{S}_{\mathcal{F}}^2[0, T]$ the set of $\mathcal{F}$ -adapted càdlàg processes $X = (X_t)_{t \in [0, T]}$ such that

$$\|X\|_{\mathbf{S}_{\mathcal{F}}^2[0, T]} = \left(\mathbb{E} \left[\sup_{t \in [0, T]} |X_t|^2 \right] \right)^{1/2} < \infty.$$

The space $\mathbf{S}_{\mathcal{F}}^2[0, T]$, endowed with this norm, is a Banach space. We also fix a compact subset A of $\mathbb{R}^q$ for some integer $q \geq 1$. An impulse strategy α consists of a double sequence $\alpha = (\tau^i, \xi^i)_{i \geq 1}$, where $(\tau^i)_{i \geq 1}$ is a non-decreasing sequence of $\mathcal{F}$ -stopping times such that $\lim_{i \rightarrow +\infty} \tau^i = +\infty$, and where ξ^i is an $\mathcal{F}_{\tau^i}$ -measurable random variable valued in A for $i \geq 1$. We define the set $\mathcal{A}$ of admissible strategies by

$$(2.1) \quad \mathcal{A} = \left\{ \alpha = (\tau^i, \xi^i)_{i \geq 1} \mid \mathbb{E} \left[\left(\sum_{i \geq 1} \mathbf{1}_{\{\tau^i \leq T\}} \right)^2 \right] < \infty \right\}.$$

We fix a drift function $b : [0, T] \times \mathbb{R}^m \rightarrow \mathbb{R}^m$, a diffusion function $\sigma : [0, T] \times \mathbb{R}^m \rightarrow \mathbb{R}^{m \times d}$, and a jump function $\gamma : \mathbb{R}^m \times A \rightarrow \mathbb{R}^m$. Given a strategy $\alpha \in \mathcal{A}$ and an initial condition $X_0 \in \mathbb{R}^m$, the controlled process X^α is defined by

(2.2)

$$X_t^\alpha = X_0 + \int_0^t b(s, X_s^\alpha) ds + \int_0^t \sigma(s, X_s^\alpha) dW_s + \sum_{0 \leq \tau^i \leq t} \gamma(X_{\tau^i-}^\alpha, \xi^i), \quad 0 \leq t \leq T.$$

At each intervention time τ^i , the process jumps according to

$$X_{\tau^i}^\alpha = X_{\tau^i-}^\alpha + \gamma(X_{\tau^i-}^\alpha, \xi^i).$$

129 If several impulses occur simultaneously, the updates are applied sequentially, i.e. if
 130 $\tau^{i+1} = \tau^i$ for some i , then, abusing the notation, we set $X_{\tau^{i+1}-}^\alpha = X_{\tau^i}^\alpha$ and therefore

$$131 \quad X_{\tau^{i+1}}^\alpha = X_{\tau^i}^\alpha + \gamma(X_{\tau^i}^\alpha, \xi^{i+1}).$$

132 Since $\lim_{i \rightarrow \infty} \tau^i = +\infty$ almost surely, only finitely many jumps occur on any finite
 133 interval, and X^α is well defined on $[0, T]$. To ensure the existence of such a controlled
 134 process, we impose the following assumption.

135 *Assumption 1.* (i) The functions b and σ are $\mathcal{B}([0, T]) \otimes \mathcal{B}(\mathbb{R}^m)$ -measurable.
 136 (ii) The function γ is $\mathcal{B}(\mathbb{R}^m) \otimes \mathcal{B}(A)$ -measurable and uniformly bounded: there
 137 exists a constant $\bar{\gamma} > 0$ such that

$$138 \quad |\gamma(x, z)| \leq \bar{\gamma}, \quad \forall (x, z) \in \mathbb{R}^m \times A.$$

139 (iii) The functions b and σ satisfy

$$140 \quad \sup_{t \in [0, T]} (|b(t, 0)| + |\sigma(t, 0)|) < \infty.$$

141 (iv) There exists a constant $L > 0$ such that, for all $(t, z, x, x') \in [0, T] \times A \times (\mathbb{R}^m)^2$,

$$142 \quad |b(t, x) - b(t, x')| + |\sigma(t, x) - \sigma(t, x')| + |\gamma(x, z) - \gamma(x', z)| \leq L|x - x'|.$$

143 **PROPOSITION 2.1.** *Under Assumption 1 there exists a unique solution X^α to 2.2*
 144 *in $\mathbf{S}_{\mathcal{F}}^2[0, T]$ for any $\alpha \in \mathcal{A}$.*

145 *Proof.* See Appendix A.1. □

146 We now fix running, terminal and impulse cost functions $f : [0, T] \times \mathbb{R}^m \rightarrow \mathbb{R}$, $g : \mathbb{R}^m \rightarrow \mathbb{R}$, $c : \mathbb{R}^m \times A \rightarrow \mathbb{R}$. We impose the following assumption.

148 *Assumption 2.* (i) The functions f , g , and c are $\mathcal{B}([0, T]) \otimes \mathcal{B}(\mathbb{R}^m)$ -, $\mathcal{B}(\mathbb{R}^m)$ -,
 149 and $\mathcal{B}(\mathbb{R}^m) \otimes \mathcal{B}(A)$ -measurable, respectively.

150 (ii) The functions f and c satisfy

$$151 \quad \sup_{t \in [0, T]} |f(t, 0)| + \sup_{z \in A} |c(0, z)| < \infty.$$

152 (iii) There exists a constant $L > 0$ such that, for all $(t, z, x, x') \in [0, T] \times A \times (\mathbb{R}^m)^2$,

$$153 \quad |g(x) - g(x')| + |f(t, x) - f(t, x')| + |c(x, z) - c(x', z)| \leq L|x - x'|.$$

154 Given an impulse strategy $\alpha \in \mathcal{A}$ and associated controlled process X^α , we define
 155 the total profit at horizon T by

$$156 \quad (2.3) \quad J(\alpha) = \mathbb{E} \left[g(X_T^\alpha) + \int_0^T f(t, X_t^\alpha) dt + \sum_{0 \leq \tau^i \leq T} c(X_{\tau^i-}^\alpha, \xi^i) \right].$$

157 Under Assumptions 1–2, the quantity $J(\alpha)$ is well defined for all $\alpha \in \mathcal{A}$. The
 158 optimal impulse control is the computation of

$$159 \quad (2.4) \quad V_0 = \sup_{\alpha \in \mathcal{A}} J(\alpha).$$

We first observe that the optimal problem over $\mathcal{A}$ may be restricted to finite strategies $\bigcup_{n \in \mathbb{N}} \mathcal{A}_n$, where

$$(2.5) \quad \mathcal{A}_n := \{ \alpha = (\tau^i, \xi^i)_{i \geq 1} \in \mathcal{A} \mid \tau^{n+1} > T \text{ a.s.} \}.$$

We then define

$$(2.6) \quad V_0^n = \sup_{\alpha \in \mathcal{A}_n} J(\alpha).$$

PROPOSITION 2.2. *Under Assumptions 1-2, we have*

$$\lim_{n \rightarrow \infty} V_0^n = V_0.$$

Proof. See Appendix A.2. □

3. Time discretization. We first introduce a structural class of admissible strategies.

Definition. Let $\delta > 0$. A strategy $\alpha = (\tau_i, \xi_i)_{i=1}^n \in \mathcal{A}_n$ is said to be δ -separated if

$$\tau_{i+1} - \tau_i \geq \delta, \quad \forall i \geq 1.$$

That is, no two impulses may occur within a time interval shorter than δ . This non-clustering property ensures the validity of short-time stability estimates required for the discretization error analysis. We now introduce a discrete-time approximation. Let $\pi = \{0 = t_0 < \dots < t_N = T\}$ be a time grid of $[0, T]$, and set

$$|\pi| = \max_{0 \leq i \leq N-1} (t_{i+1} - t_i).$$

We define the corresponding discretely admissible strategies as

$$\mathcal{A}_{n,\pi} := \left\{ \alpha = (\tau^i, \xi^i)_{1 \leq i \leq n} : \tau^i \in \pi \cup \{T+1\}, \tau^1 \leq \dots \leq \tau^n, \xi^i \in L^0(\mathcal{F}_{\tau^i}; A) \right\}.$$

$$\mathcal{A}_{n,\delta} := \left\{ \alpha = (\tau^i, \xi^i)_{1 \leq i \leq n} : \tau^i \in \pi \cup \{T+1\}, \tau^{i+1} - \tau^i \geq \delta, \xi^i \in L^0(\mathcal{F}_{\tau^i}; A) \right\},$$

and we denote

$$V_0^{n,\pi} := \sup_{\alpha \in \mathcal{A}_{n,\pi}} J(\alpha).$$

$$V_0^{n,\delta} := \sup_{\alpha \in \mathcal{A}_{n,\delta}} J(\alpha).$$

The next proposition shows that any δ -separated strategy can be approximated by a grid-based strategy with an error of order $O(|\pi|^{1/2})$.

PROPOSITION 3.1. *Assume Assumptions 1-2 and fix $n \in \mathbb{N}$. Let $\alpha = (\tau_i, \xi_i)_{i=1}^n \in \mathcal{A}_n$ be a δ -separated strategy for some $\delta > 0$. Then there exists a constant $\Lambda > 0$, depending on $L, K, T, \bar{\gamma}, n$ and $\mathbb{E}[|X_0|^2]$, such that for every time grid π satisfying $|\pi| \leq \delta$, there exists a strategy $\alpha^\pi \in \mathcal{A}_{n,\pi}$ such that*

$$|J(\alpha) - J(\alpha^\pi)| \leq \Lambda |\pi|^{1/2}.$$

As an immediate consequence, for every time grid π satisfying $|\pi| \leq \delta$,

$$|V_0^{n,\delta} - V_0^{n,\pi}| \leq \Lambda |\pi|^{1/2},$$

In particular, if the impulse control problem is restricted to δ -separated strategies for some fixed $\delta > 0$, then the corresponding discrete-time approximation converges to the restricted value function as the mesh size tends to zero.

In order to prove Proposition 3.1, we begin by establishing a set of auxiliary estimates. Lemmas 3.2 and 3.3 correspond to classical short-time increment bounds of [46, Theorems 1.2 and 1.4], here adapted to the setting of bounded stopping times. Their proofs rely on the same arguments as in the deterministic-time case, and for this reason are not reproduced here.

LEMMA 3.2. *Assume Assumption 1, so that in particular b and σ are globally Lipschitz in x with constant $L > 0$ and satisfy the linear growth condition*

$$|b(t, x)|^2 + \|\sigma(t, x)\|^2 \leq K(1 + |x|^2), \quad (t, x) \in [0, T] \times \mathbb{R}^m.$$

Let X be a càdlàg process and let τ_1, τ_2 be $\mathcal{F}$ -bounded stopping times such that

$$\tau_1 \leq \tau_2, \quad \tau_2 - \tau_1 \leq |\pi|,$$

for some $|\pi| \in (0, T]$. Assume that, on the random interval $[\tau_1, \tau_2)$, the process X has no jump and satisfies

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t.$$

Then there exists a constant $\Lambda > 0$, depending only on L, K and T , such that

$$\mathbb{E} \left[\sup_{\tau_1 \leq u < \tau_2} |X_u - X_{\tau_1}|^2 \right] \leq \Lambda |\pi| \left(1 + \mathbb{E} |X_{\tau_1}|^2 \right) e^{\Lambda |\pi|}.$$

LEMMA 3.3. *Assume Assumption 1. Let τ_1, τ_2 be $\mathcal{F}$ -bounded stopping times such that*

$$\tau_1 \leq \tau_2 \leq T, \quad \tau_2 - \tau_1 \leq |\pi|,$$

for some $|\pi| \in (0, T]$. Assume that, on the random interval $[\tau_1, \tau_2)$, the processes $X^{(1)}$ and $X^{(2)}$ have no jump and are two solutions of the SDE

$$dX_t^{(i)} = b(t, X_t^{(i)}) dt + \sigma(t, X_t^{(i)}) dW_t, \quad i = 1, 2,$$

Then there exists a constant $\Lambda > 0$, depending only on L, K, T , such that

$$\mathbb{E} \left[\sup_{\tau_1 \leq t < \tau_2} \left| (X_t^{(1)} - X_t^{(2)}) - (X_{\tau_1}^{(1)} - X_{\tau_1}^{(2)}) \right|^2 \right] \leq \Lambda |\pi| \mathbb{E} \left[|X_{\tau_1}^{(1)} - X_{\tau_1}^{(2)}|^2 \right] e^{\Lambda |\pi|}.$$

We also require a uniform moment bound for the controlled state process over the entire class $\mathcal{A}_n$. Such an estimate will be used repeatedly in the proof of Proposition 3.1 to ensure that all constants remain independent of the particular strategy chosen within $\mathcal{A}_n$. The following lemma provides precisely this uniform control.

LEMMA 3.4. *Assume Assumption 1 and fix $n \in \mathbb{N}$. There exists a constant $\Lambda > 0$, depending only on $L, K, T, \bar{\gamma}, n$ and $\mathbb{E}[|X_0|^2]$, such that*

$$\sup_{\alpha \in \mathcal{A}_n} \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^\alpha|^2 \right] \leq \Lambda.$$

227 *Proof.* See Appendix A.3. □

228 We are now ready to prove Proposition 3.1.

229 *Proof of Proposition 3.1.* Fix a time grid $\pi = \{0 = t_0 < t_1 < \dots < t_N = T\}$ such
 230 that $|\pi| \leq \delta$. Define the discretization of α on π by $\alpha^\pi := (\tau_k^\pi, \xi_k^\pi)_{1 \leq k \leq n}$ where

$$231 \quad \tau_k^\pi := \inf\{t_j \in \pi : t_j \geq \tau_k\}, \quad \xi_k^\pi := \xi_k,$$

232 and denote $X := X^\alpha$ and $X^\alpha := X^{\alpha^\pi}$ the associated controlled processes.

233 Since α is δ -separated and $|\pi| \leq \delta$, every interval $[\tau_k, \tau_k^\pi)$ has length at most $|\pi|$
 234 and cannot contain any additional intervention time of α . By Lemma 3.4, for each n
 235 we denote by $\Lambda_{(n)} > 0$ the constant such that

$$236 \quad \sup_{\alpha \in \mathcal{A}_n} \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^\alpha|^2 \right] \leq \Lambda_{(n)}.$$

237 **Step 1. Error decomposition.** By definition of the cost functionals,

$$238 \quad J(\alpha^\pi) - J(\alpha) = \mathbb{E} \left[g(X_T^\alpha) - g(X_T) + \int_0^T (f(t, X_t^\alpha) - f(t, X_t)) dt \right. \\
 239 \quad \left. + \sum_{k=1}^n (c(X_{\tau_k^\pi -}^\alpha, \xi_k) - c(X_{\tau_k -}, \xi_k)) \right].$$

240 By Lipschitz continuity of f, g, c (with constant L),

$$241 \quad (3.1) \quad |J(\alpha^\pi) - J(\alpha)| \leq L \mathbb{E} \left[|X_T^\alpha - X_T| + \int_0^T |X_t^\alpha - X_t| dt + \sum_{k=1}^n |X_{\tau_k^\pi -}^\alpha - X_{\tau_k -}| \right].$$

242 For each $k = 1, \dots, n$, set

$$243 \quad \bar{I}_k := [\tau_{k-1}^\pi, \tau_k), \quad I_k := [\tau_k, \tau_k^\pi), \quad \tau_0^\pi := 0.$$

244 By construction, $\tau_k^\pi - \tau_k \leq |\pi|$ for all k . We claim that there exists $\Lambda > 0$, independent
 245 of $|\pi|$, such that for all $k = 1, \dots, n$,

$$246 \quad (e_k^1) \quad \mathbb{E} \left[\int_{\bar{I}_k} |X_t^\alpha - X_t|^2 dt \right] \leq \Lambda |\pi|,$$

$$247 \quad (e_k^2) \quad \mathbb{E} [|X_{\tau_k^\pi -}^\alpha - X_{\tau_k -}|^2] \leq \Lambda |\pi|,$$

$$248 \quad (e_k^3) \quad \mathbb{E} [|X_{\tau_k^\pi}^\alpha - X_{\tau_k^\pi}^\pi|^2] \leq \Lambda |\pi|,$$

$$249 \quad (e_k^4) \quad \mathbb{E} \left[\int_{I_k} |X_t^\alpha - X_t|^2 dt \right] \leq \Lambda |\pi|$$

250 **Step 2. Induction.**

251 *Case $k = 1$.*

252 (e_1^1) : On $\bar{I}_1$ no impulse occurs so the claim follows.

253 (e_1^2) : Since $X_{\tau_1 -}^\alpha = X_{\tau_1 -}$, we have

$$254 \quad \mathbb{E} [|X_{\tau_1^\pi -}^\alpha - X_{\tau_1 -}|^2] = \mathbb{E} [|X_{\tau_1^\pi}^\alpha - X_{\tau_1}^\alpha|^2].$$

and the claim follows by Lemma 3.2 and 3.4

(e_1^3) : We have

$$\begin{aligned}
\mathbb{E}[|X_{\tau_1^\pi}^\alpha - X_{\tau_1}^\pi|^2] &\leq \mathbb{E}\left[|(X_{\tau_1^\pi}^\alpha + \gamma(X_{\tau_1^\pi}^\alpha, \xi_1)) - X_{\tau_1^\pi} + X_{\tau_1} - X_{\tau_1}|^2\right] \\
&\leq \mathbb{E}\left[|(X_{\tau_1^\pi}^\alpha - X_{\tau_1}) + (X_{\tau_1^\pi} - X_{\tau_1})\right. \\
&\quad \left.+ (\gamma(X_{\tau_1^\pi}^\alpha, \xi_1) - \gamma(X_{\tau_1}, \xi_1))|^2\right] \\
&\leq 3\mathbb{E}[|X_{\tau_1^\pi}^\alpha - X_{\tau_1}|^2] + 3\mathbb{E}[|X_{\tau_1^\pi} - X_{\tau_1}|^2] \\
&\quad + 3\mathbb{E}[|\gamma(X_{\tau_1^\pi}^\alpha, \xi_1) - \gamma(X_{\tau_1}, \xi_1)|^2] \\
&\leq 3(1+L)\mathbb{E}[|X_{\tau_1^\pi}^\alpha - X_{\tau_1}|^2] \\
&\quad + 3\mathbb{E}[|X_{\tau_1^\pi} - X_{\tau_1}|^2].
\end{aligned}$$

where we use the Lipschitz property of γ in the fourth line, Lemma 3.2 and estimate (e_1^2) in the fifth line, and Lemma 3.4 in the sixth line. This proves (e_1^3) .

(e_1^4) : We have

$$\begin{aligned}
\mathbb{E}\left[\int_{\tau_1}^{\tau_1^\pi} |X_t^\alpha - X_t|^2 dt\right] &\leq 2\mathbb{E}\left[\int_{\tau_1}^{\tau_1^\pi} |(X_t^\alpha - X_t) - (X_{\tau_1}^\alpha - X_{\tau_1})|^2 dt\right] \\
&\quad + 2\mathbb{E}\left[\int_{\tau_1}^{\tau_1^\pi} |X_{\tau_1}^\alpha - X_{\tau_1}|^2 dt\right] \\
&\leq 2(\Lambda_1|\pi|^2 + |\pi|)\mathbb{E}[|X_{\tau_1}^\alpha - X_{\tau_1}|^2] \\
&\leq 4(\Lambda_1|\pi|^2 + |\pi|)\Lambda(n) \\
&\leq \Lambda|\pi|.
\end{aligned}$$

using Lemma 3.3 and estimate (e_1^3) . This proves (e_1^4) .

General case $k \geq 2$.

(e_k^1) : estimate on $\bar{I}_k = [\tau_{k-1}^\pi, \tau_k)$.

$$\begin{aligned}
\mathbb{E}\left[\int_{\tau_{k-1}^\pi}^{\tau_k} |X_t^\alpha - X_t|^2 dt\right] &\leq 2\mathbb{E}\left[\int_{\tau_{k-1}^\pi}^{\tau_k} |(X_t^\alpha - X_t) - (X_{\tau_{k-1}^\pi}^\alpha - X_{\tau_{k-1}^\pi})|^2 dt\right] \\
&\quad + 2\mathbb{E}\left[\int_{\tau_{k-1}^\pi}^{\tau_k} |X_{\tau_{k-1}^\pi}^\alpha - X_{\tau_{k-1}^\pi}|^2 dt\right] \\
&\leq 2(\Lambda_1 T^2 + T)\mathbb{E}[|X_{\tau_{k-1}^\pi}^\alpha - X_{\tau_{k-1}^\pi}|^2] \\
&\leq \Lambda|\pi|,
\end{aligned}$$

using Lemma 3.3 and (e_{k-1}^3) . This proves e_k^1 .

280 (e_k^2) : First,

$$\begin{aligned}
 281 \quad \mathbb{E}[|X_{\tau_k-}^\alpha - X_{\tau_k-}|^2] &\leq 2\mathbb{E}\left[|(X_{\tau_k-}^\alpha - X_{\tau_k-}) - (X_{\tau_{k-1}^\pi}^\alpha - X_{\tau_{k-1}^\pi})|^2\right] \\
 282 \quad &\quad + 2\mathbb{E}\left[|X_{\tau_{k-1}^\pi}^\alpha - X_{\tau_{k-1}^\pi}|^2\right] \\
 283 \quad &\leq 2(\Lambda_1 T + 1)\mathbb{E}\left[|X_{\tau_{k-1}^\pi}^\alpha - X_{\tau_{k-1}^\pi}|^2\right] \\
 284 \quad &\leq \Lambda|\pi|.
 \end{aligned}$$

285 where we use Lemma 3.3 and (e_{k-1}^3) . Now we write

$$\begin{aligned}
 286 \quad \mathbb{E}[|X_{\tau_k^\pi}^\alpha - X_{\tau_k-}|^2] &\leq \mathbb{E}[|X_{\tau_k^\pi}^\alpha - X_{\tau_k-}^\alpha + X_{\tau_k-}^\alpha - X_{\tau_k-}|^2] \\
 287 \quad &\leq 2\mathbb{E}[|X_{\tau_k^\pi}^\alpha - X_{\tau_k-}^\alpha|^2] + 2\mathbb{E}[|X_{\tau_k-}^\alpha - X_{\tau_k-}|^2] \\
 288 \quad &\leq 2\Lambda_2|\pi|(1 + \mathbb{E}[|X_{\tau_k}^\alpha|^2]) + 2\mathbb{E}[|X_{\tau_k-}^\alpha - X_{\tau_k-}|^2] \\
 289 \quad &\leq 2\Lambda_2|\pi|(1 + \Lambda(n)) + 2\mathbb{E}[|X_{\tau_k-}^\alpha - X_{\tau_k-}|^2] \\
 290 \quad &\leq \Lambda|\pi|,
 \end{aligned}$$

291 using Lemma 3.2 and the inequality above. This proves e_k^2 .

292 (e_k^3) : The same calculation as in $(e_1^3) - (e_1^3)$ yields the desired result, using (e_k^3) ,
 293 Lemma 3.2, and Lemma 3.4.

294 **Step 3. Conclusion.** Summing $e_k^1 - e_k^4$ over $k = 1, \dots, n$ gives

$$295 \quad \mathbb{E}\left[\int_0^{\tau_n^\pi} |X_t^\alpha - X_t|^2 dt\right] + \sum_{k=1}^n \mathbb{E}[|X_{\tau_k^\pi}^\alpha - X_{\tau_k-}|^2] \leq \Lambda|\pi|.$$

296 On the interval $[\tau_n^\pi, T]$ no impulse occurs, so the same argument used in the recursion
 297 yields

$$298 \quad \mathbb{E}\left[\int_{\tau_n^\pi}^T |X_t^\alpha - X_t|^2 dt\right] + \mathbb{E}[|X_T^\alpha - X_T|^2] \leq \Lambda|\pi|.$$

299 Hence

$$300 \quad \mathbb{E}\left[\int_0^T |X_t^\alpha - X_t|^2 dt\right] + \mathbb{E}[|X_T^\alpha - X_T|^2] + \sum_{k=1}^n \mathbb{E}[|X_{\tau_k^\pi}^\alpha - X_{\tau_k-}|^2] \leq \Lambda|\pi|.$$

301 By Cauchy-Schwarz,

$$302 \quad \mathbb{E}\left[\int_0^T |X_t^\alpha - X_t| dt\right] + \mathbb{E}[|X_T^\alpha - X_T|] + \sum_{k=1}^n \mathbb{E}[|X_{\tau_k^\pi}^\alpha - X_{\tau_k-}|] \leq \Lambda|\pi|^{1/2}.$$

303 Inserting these bounds into 3.1 yields

$$304 \quad |J(\alpha^\pi) - J(\alpha)| \leq \Lambda|\pi|^{1/2},$$

305 for some constant $\Lambda > 0$ depending only on $L, K, T, \bar{\gamma}, n$ and $\mathbb{E}[|X_0|^2]$. $\square$

4. Dynamic programming and solution on a finite grid. We stay in the discrete-time impulse control problem obtained by restricting intervention times to a finite grid $\{0 = t_0 < t_1 < \dots < t_K = T\}$. At most one impulse can occur at each date t_k , so that only finitely many interventions are possible. This allows us to fully exploit the dynamic programming principle on the grid and to construct an optimal strategy recursively from the comparison between continuation and intervention values.

Here an impulse strategy is a sequence $\{(\tau_n, \xi_n)\}_{n \geq 1}$ where

- each τ_n is an $(\mathcal{F}_t)$ -stopping time taking values in $\{t_0, \dots, t_K, T + 1\}$, where $T + 1$ denotes a cemetery time,
- $\tau_1 < \tau_2 < \dots \leq T + 1$ almost surely,
- ξ_n is $\mathcal{F}_{\tau_n}$ -measurable and takes values in $D(X_{\tau_n-})$, where $D(x) \subset \mathbb{R}^q$ denotes the compact set of admissible impulse controls when the state is x .

For $k \in \{0, \dots, K\}$, we let

$$\Pi_k := \{(\tau_n, \xi_n)_{n \geq 1} : \tau_1 \geq t_k\},$$

the set of strategies whose first impulse occurs no earlier than t_k . Given $k \in \{0, \dots, K\}$ and an initial condition $x \in \mathbb{R}^m$, the dynamics under a strategy $\alpha \in \Pi_k$ is

$$\begin{aligned} X_t^{k,x,\alpha} &= x + \int_{t_k}^t b(s, X_s^{k,x,\alpha}) ds + \int_{t_k}^t \sigma(s, X_s^{k,x,\alpha}) dW_s \\ &\quad + \sum_{\tau_n \in [t_k, t]} \gamma(X_{\tau_n-}^{k,x,\alpha}, \xi_n), \quad t \in [t_k, T]. \end{aligned}$$

When no impulse is applied, the state simply follows the uncontrolled diffusion,

$$X_t^{k,x,\emptyset} = x + \int_{t_k}^t b(s, X_s^{k,x,\emptyset}) ds + \int_{t_k}^t \sigma(s, X_s^{k,x,\emptyset}) dW_s, \quad t \in [t_k, T].$$

At time t_k , the performance of a strategy $\alpha \in \Pi_k$ is measured through the total discounted cost accumulated over $[t_k, T]$:

$$\begin{aligned} J_k(x; \alpha) &:= \mathbb{E} \left[\int_{t_k}^T e^{-\rho(s-t_k)} f(s, X_s^{k,x,\alpha}) ds \right. \\ &\quad \left. + \sum_{\substack{(\tau, \xi) \in \alpha \\ \tau \in [t_k, T]}} e^{-\rho(\tau-t_k)} c(X_{\tau-}^{k,x,\alpha}, \xi) + e^{-\rho(t_K-t_k)} \phi(X_{t_K}^{k,x,\alpha}) \right]. \end{aligned}$$

The associated value function is obtained by optimizing over all strategies that intervene no earlier than t_k :

$$V_k(x) := \sup_{\alpha \in \Pi_k} J_k(x; \alpha).$$

The dynamic programming iteration is formulated in terms of two operators. At the terminal time $t_K = T$, continuation yields the terminal cost

$$C_K := \phi.$$

At times t_k with $k < K$, the continuation value is the expected cost incurred from t_k to t_{k+1} under the uncontrolled dynamics, plus the discounted value function at the next grid point:

$$C_k(x) := \mathbb{E} \left[\int_{t_k}^{t_{k+1}} e^{-\rho(s-t_k)} f(s, X_s^{k,x,\emptyset}) ds + e^{-\rho(t_{k+1}-t_k)} V_{k+1}(X_{t_{k+1}}^{k,x,\emptyset}) \right].$$

In parallel, at each time t_k we may apply an impulse $\xi \in D(x)$, incurring an immediate cost and jumping to the post-intervention state $x + \gamma(x, \xi)$. The corresponding intervention operator is therefore

$$(4.1) \quad M_k(x) := \sup_{\xi \in D(x)} \{c(x, \xi) + C_k(x + \gamma(x, \xi))\}.$$

We assume that measurable selectors $\xi_k : E \rightarrow \mathbb{R}^m$ exist such that

$$M_k(x) = c(x, \xi_k(x)) + C_k(x + \gamma(x, \xi_k(x))), \quad x \in E.$$

Together, C_k and M_k fully characterize the optimal behaviour at time t_k . The dynamic programming principle states that the controller chooses whichever action yields the higher reward:

$$V_k(x) = \max\{C_k(x), M_k(x)\}, \quad k = 0, \dots, K, \quad x \in \mathbb{R}^m.$$

The comparison between C_k and M_k naturally induces a partition of the state space at each grid time. Whenever $M_k(x) > C_k(x)$, applying an impulse is preferable to continuation, and we therefore introduce the intervention region

$$\mathcal{I}_k := \{x \in E : M_k(x) > C_k(x)\}.$$

This decomposition lies at the core of the optimal policy: starting from (t_k, x) , the controller simply checks whether the state enters $\mathcal{I}_k$ (and, if so, at which grid point) to determine if the intervention should occur at t_k or later.

To formalize this idea, consider the uncontrolled trajectory $X^{k,x,\emptyset}$ on $[t_k, T]$. We define two intervention times. The first one records the earliest grid time $t_j \geq t_k$ at which the state hits the appropriate intervention region,

$$\tau(t_k, x) := \inf\{t_j \geq t_k : X_{t_j}^{k,x,\emptyset} \in \mathcal{I}_j\},$$

while the second enforces a strict delay past t_k ,

$$\tilde{\tau}(t_k, x) := \inf\{t_j > t_k : X_{t_j}^{k,x,\emptyset} \in \mathcal{I}_j\}.$$

If no intervention becomes optimal over the relevant portion of the grid, both quantities are set to the cemetery time $T + 1$ so we set by convention $\inf\{\emptyset\} = T + 1$. In turn, the impulse actually applied at these times is determined through the selector ξ_j :

$$\xi(t_k, x) := \begin{cases} \xi_j(x), & \text{if } \tau(t_k, x) = t_j, \\ 0, & \text{if } \tau(t_k, x) = T + 1, \end{cases} \quad \tilde{\xi}(t_k, x) := \begin{cases} \xi_j(x), & \text{if } \tilde{\tau}(t_k, x) = t_j, \\ 0, & \text{if } \tilde{\tau}(t_k, x) = T + 1. \end{cases}$$

The operators τ and $\tilde{\tau}$ play two complementary roles. The map $\tau(t_k, x)$ records the first grid time $t_j \geq t_k$ at which it becomes optimal to intervene when impulses at t_k are allowed. It is the basic building block for constructing, step by step, an optimal strategy that may act immediately at the current time. In contrast, $\tilde{\tau}(t_k, x)$ enforces inaction at t_k and only detects the first optimal intervention time strictly after t_k . The pair $(\tilde{\tau}, \tilde{\xi})$ therefore describes the best possible behaviour under the explicit constraint of not acting at t_k . This viewpoint is closely related to the dynamic programming decomposition: the continuation value $C_k(x)$ will be realized by the performance of a

strategy that is forced to wait at t_k , whereas the full value $V_k(x)$ will correspond to the performance of a strategy that is allowed to apply an immediate impulse whenever this is beneficial.

Having introduced these operators, we now detail the construction of two strategies $\alpha^*(t_k, x)$ and $\tilde{\alpha}^*(t_k, x)$. A verification lemma will then show that these two strategies realize, respectively, the value function $V_k(x)$ and the continuation value $C_k(x)$.

Construction of an optimal strategy. The operators (τ, ξ) and $(\tilde{\tau}, \tilde{\xi})$ allow us to construct iteratively, from any starting pair (t_k, x) , a strategy

$$\alpha^*(t_k, x) := \{(\tau_i^{k,x}, \xi_i^{k,x})\}_{i \geq 1}$$

solely from the uncontrolled dynamics and the intervention regions.

Initialization. The first intervention is obtained exactly as in the one-step rule:

$$\tau_1^{k,x} := \tau(t_k, x), \quad \xi_1^{k,x} := \xi(t_k, x).$$

Thus an impulse may occur immediately when $x \in \mathcal{I}_k$, or later along the grid when $x \notin \mathcal{I}_k$.

Induction step. Assume that $(\tau_1^{k,x}, \dots, \tau_{n-1}^{k,x})$ and $(\xi_1^{k,x}, \dots, \xi_{n-1}^{k,x})$ have been constructed. We denote by

$$\alpha^{*,(n-1)}(t_k, x)$$

the strategy consisting of the first $n-1$ intervention times already constructed, followed by no further control. Let $X^{k,x,\alpha^{*,(n-1)}}$ be the associated controlled process. At time $\tau_{n-1}^{k,x}$, the current post-impulse state is $X_{\tau_{n-1}^{k,x}}^{k,x,\alpha^{*,(n-1)}}$. The next intervention is obtained by applying exactly the same rule as for the first impulse, but starting from this updated pair:

$$\tau_n^{k,x} := \tilde{\tau}\left(\tau_{n-1}^{k,x}, X_{\tau_{n-1}^{k,x}}^{k,x,\alpha^{*,(n-1)}}\right), \quad \xi_n^{k,x} := \tilde{\xi}\left(\tau_{n-1}^{k,x}, X_{\tau_{n-1}^{k,x}}^{k,x,\alpha^{*,(n-1)}}\right).$$

This enforces a strict delay after the previous intervention while inheriting the same intervention rule. Iterating this construction yields a nondecreasing sequence of stopping times $(\tau_n^{k,x})_{n \geq 1} \subset \{t_k, \dots, t_K, T+1\}$ together with the corresponding sequence of impulse magnitudes $(\xi_n^{k,x})_{n \geq 1}$.

Collecting all impulses produced by this iterative rule gives the canonical strategy $\alpha^*(t_k, x)$. It is entirely determined by the uncontrolled dynamics and the geometry of the intervention regions. For later reference, we also introduce the variant $\tilde{\alpha}^*(t_k, x)$, built in exactly the same way except that the first step uses $(\tilde{\tau}, \tilde{\xi})$ instead of (τ, ξ) , i.e. no impulse is allowed at the initial time t_k .

A direct inspection of the above construction shows that

$$\tilde{\alpha}^*(t_k, x) = \alpha^*(t_{k+1}, X_{t_{k+1}}^{k,x,\emptyset}),$$

with the convention that the strategy is empty once $T+1$ is reached. Moreover,

- If $x \in \mathcal{I}_k$, then the strategy intervenes immediately at time t_k with impulse $\xi_k(x)$, producing the post-impulse state

$$x' := x + \gamma(x, \xi_k(x)).$$

From that point on, the strategy simply follows $\tilde{\alpha}^*(t_k, x')$, so that $\alpha^*(t_k, x)$ is obtained by placing the intervention $(t_k, \xi_k(x))$ in front of the sequence $\tilde{\alpha}^*(t_k, x')$.

- If $x \notin \mathcal{I}_k$, then no intervention is taken at time t_k , and the entire sequence is simply inherited from the next grid point:

$$\alpha^*(t_k, x) = \alpha^*(t_{k+1}, X_{t_{k+1}}^{k,x,\emptyset}).$$

LEMMA 4.1 (Verification). *For each $k \in \{0, \dots, K\}$ and each $x \in E$, the strategies $\alpha^*(t_k, x)$ and $\tilde{\alpha}^*(t_k, x)$ satisfy*

$$J_k(x; \alpha^*(t_k, x)) = V_k(x) \quad \text{and} \quad J_k(x; \tilde{\alpha}^*(t_k, x)) = C_k(x).$$

In particular, $\alpha^*(t_k, x)$ is an optimal impulse strategy on $[t_k, T]$.

Proof. We argue by backward induction on k .

Step $k = K$. At time $t_K = T$ there is no continuation, so $C_K = \phi$ and

$$V_K(x) = \max\{\phi(x), M_K(x)\}, \quad x \in E.$$

By definition, $\tilde{\alpha}^*(t_K, x)$ never intervenes at t_K , hence

$$J_K(x; \tilde{\alpha}^*(t_K, x)) = \phi(x) = C_K(x).$$

The strategy $\alpha^*(t_K, x)$ either does nothing (when $x \notin \mathcal{I}_K$) or applies the impulse $\xi_K(x)$ at time T (when $x \in \mathcal{I}_K$), so that

$$J_K(x; \alpha^*(t_K, x)) = \begin{cases} \phi(x), & x \notin \mathcal{I}_K, \\ M_K(x), & x \in \mathcal{I}_K, \end{cases}$$

and therefore $J_K(x; \alpha^*(t_K, x)) = V_K(x)$.

Induction step. Fix $k \in \{0, \dots, K-1\}$ and assume that, for every $y \in E$,

$$J_{k+1}(y; \alpha^*(t_{k+1}, y)) = V_{k+1}(y), \quad J_{k+1}(y; \tilde{\alpha}^*(t_{k+1}, y)) = C_{k+1}(y).$$

1) *The strategy $\tilde{\alpha}^*(t_k, x)$.* By construction, $\tilde{\alpha}^*(t_k, x)$ takes no impulse at time t_k and, from t_{k+1} onward, coincides with $\alpha^*(t_{k+1}, \cdot)$ applied to the uncontrolled state $X_{t_{k+1}}^{k,x,\emptyset}$. Hence

$$\begin{aligned} J_k(x; \tilde{\alpha}^*(t_k, x)) &= \mathbb{E} \left[\int_{t_k}^{t_{k+1}} e^{-\rho(s-t_k)} f(X_s^{k,x,\emptyset}) ds \right. \\ &\quad \left. + e^{-\rho(t_{k+1}-t_k)} J_{k+1}(X_{t_{k+1}}^{k,x,\emptyset}; \alpha^*(t_{k+1}, X_{t_{k+1}}^{k,x,\emptyset})) \right]. \end{aligned}$$

Using the induction hypothesis and the definition of C_k yields

$$\begin{aligned} J_k(x; \tilde{\alpha}^*(t_k, x)) &= \mathbb{E} \left[\int_{t_k}^{t_{k+1}} e^{-\rho(s-t_k)} f(X_s^{k,x,\emptyset}) ds \right. \\ &\quad \left. + e^{-\rho(t_{k+1}-t_k)} V_{k+1}(X_{t_{k+1}}^{k,x,\emptyset}) \right] = C_k(x). \end{aligned}$$

2) *The strategy $\alpha^*(t_k, x)$.* We distinguish two cases according to the intervention region $\mathcal{I}_k = \{M_k > C_k\}$.

If $x \notin \mathcal{I}_k$, then $\alpha^*(t_k, x)$ does not intervene at time t_k and, by definition of $\tilde{\alpha}^*$, we have

$$\alpha^*(t_k, x) = \tilde{\alpha}^*(t_k, x).$$

It follows from the first part that

$$J_k(x; \alpha^*(t_k, x)) = J_k(x; \tilde{\alpha}^*(t_k, x)) = C_k(x).$$

If $x \in \mathcal{I}_k$, then $\alpha^*(t_k, x)$ applies the impulse $\xi_k(x)$ at time t_k . Let $x' := x + \gamma(x, \xi_k(x))$ denote the post-impulse state. Between t_k and t_{k+1} the dynamics is uncontrolled, and from t_{k+1} onward the strategy coincides with $\alpha^*(t_{k+1}, \cdot)$ started from $X_{t_{k+1}}^{t_k, x', \emptyset}$. Therefore

$$\begin{aligned} J_k(x; \alpha^*(t_k, x)) &= c(x, \xi_k(x)) + \mathbb{E} \left[\int_{t_k}^{t_{k+1}} e^{-\rho(s-t_k)} f(X_s^{t_k, x', \emptyset}) ds \right. \\ &\quad \left. + e^{-\rho(t_{k+1}-t_k)} J_{k+1}(X_{t_{k+1}}^{t_k, x', \emptyset}; \alpha^*(t_{k+1}, X_{t_{k+1}}^{t_k, x', \emptyset})) \right]. \end{aligned}$$

Applying the induction hypothesis at time t_{k+1} and the definition of C_k gives

$$J_k(x; \alpha^*(t_k, x)) = c(x, \xi_k(x)) + C_k(x + \gamma(x, \xi_k(x))).$$

By definition of M_k and the selector ξ_k , the right-hand side is precisely $M_k(x)$. Since $x \in \mathcal{I}_k$ we have $M_k(x) > C_k(x)$, so

$$V_k(x) = \max\{C_k(x), M_k(x)\} = M_k(x) = J_k(x; \alpha^*(t_k, x)).$$

Combining the two cases completes the induction and the proof. $\square$

5. Algorithm. Section 4 together with Appendix B provide the main ingredients for constructing an algorithm that approximates the solution of the problem. In particular, the strategy $\tilde{\alpha}^*(t_k, x)$ constructed in Section 4 plays a central role. It corresponds to the strategy that, starting from (t_k, x) , forbids any impulse at time t_k and then follows the intervention rule induced by the regions $(\mathcal{I}_j)_{j \geq k+1}$.

For each $k \in \{0, \dots, K-1\}$ and $x \in E$, we denote by

$$X_s^{k, x, \tilde{\alpha}^*} := X_s^{k, x, \tilde{\alpha}^*(t_k, x)}$$

the corresponding controlled state process, and define the associated continuation cost

$$\begin{aligned} Y^{k, x} &= \int_{t_k}^T e^{-\rho(s-t_k)} f(X_s^{k, x, \tilde{\alpha}^*}) ds \\ &\quad + \sum_{(\tau, \xi) \in \tilde{\alpha}^*(t_k, x), \tau \in [t_k, T]} e^{-\rho(\tau-t_k)} c(X_{\tau-}^{k, x, \tilde{\alpha}^*}, \xi) + e^{-\rho(t_K-t_k)} \phi(X_{t_K}^{k, x, \tilde{\alpha}^*}). \end{aligned}$$

Lemma 4.1 implies that $\tilde{\alpha}^*(t_k, x)$ realizes the continuation value at time t_k so

$$\mathbb{E}[Y^{k, x}] = J_k(x; \tilde{\alpha}^*(t_k, x)) = C_k(x), \quad k = 0, \dots, K-1, x \in E.$$

Thus C_k can be obtained as the expectation of the random variable $Y^{k, x}$, whose law is determined by the controlled dynamics under $\tilde{\alpha}^*(t_k, x)$. Crucially, once the operators $(C_{k+1}, \dots, C_K)$ and the selectors $(\xi_{k+1}, \dots, \xi_K)$ are available, the path $(X^{k, x, \tilde{\alpha}^*}, Y^{k, x})$ can be generated using Algorithm 5.1. In this way, the computation of each C_k relies on simulations driven by $(C_{k+1}, \dots, C_K)$, leading to a backward alternation between simulation and regression. The resulting numerical scheme is summarized in Algorithm 5.2 which is mostly inspired by the work of Ludkovski [39].

Algorithm 5.1 Simulation of $Y^{k,x}$

1. Set $X_{t_k} = x$ and $Y = 0$.
2. Simulate the uncontrolled diffusion on $[t_k, t_{k+1}]$ and update

$$Y \leftarrow Y + \int_{t_k}^{t_{k+1}} e^{-\rho(s-t_k)} f(X_s) ds.$$

3. For $j = k+1, \dots, K$:
 - (a) Compute $C_j(X_{t_j})$ and $M_j(X_{t_j})$.
 - (b) If $M_j(X_{t_j}) > C_j(X_{t_j})$, update

$$Y \leftarrow Y + e^{-\rho(t_j-t_k)} c(X_{t_j}, \xi_j(X_{t_j}))$$

and

$$X_{t_j} \leftarrow X_{t_j} + \gamma(X_{t_j}, \xi_j(X_{t_j})).$$

- (c) If $j < K$, simulate the uncontrolled diffusion on $[t_j, t_{j+1}]$ and update

$$Y \leftarrow Y + \int_{t_j}^{t_{j+1}} e^{-\rho(s-t_k)} f(X_s) ds.$$

4. Set

$$Y \leftarrow Y + e^{-\rho(t_K-t_k)} \phi(X_{t_K}).$$

5. Return Y .

5.1. Finite number of impulses. We extend the algorithms to the case where we constrain the controller to perform at most $M \geq 0$ interventions over $[0, T]$. For each $k \in \{0, \dots, K\}$ and each $m \in \{0, \dots, M\}$, let $V_{k,m}(x)$ denote the optimal value starting from (t_k, x) with m impulses remaining. The backward equations become

$$C_{k,m}(x) = \mathbb{E} \left[\int_{t_k}^{t_{k+1}} e^{-\rho(s-t_k)} f(X_s^{k,x,\emptyset}) ds + e^{-\rho(t_{k+1}-t_k)} V_{k+1,m}(X_{t_{k+1}}^{k,x,\emptyset}) \right],$$

meaning that absence of intervention at t_k preserves the remaining budget m . When $m \geq 1$, a single intervention at t_k consumes one unit of budget and must be followed by continuation at t_k :

$$M_{k,m}(x) := \sup_{\xi \in D(x)} \left\{ c(x, \xi) + C_{k,m-1}(x + \gamma(x, \xi)) \right\}.$$

The value function is then

$$V_{k,m}(x) = \max\{C_{k,m}(x), M_{k,m}(x)\}.$$

At maturity,

$$C_{K,m}(x) = \phi(x), \quad m = 0, \dots, M.$$

With no impulses left, intervention becomes impossible and the value reduces to the uncontrolled cost:

$$V_{k,0}(x) = C_{k,0}(x) = \mathbb{E} \left[\int_{t_k}^T e^{-\rho(s-t_k)} f(X_s^{k,x,\emptyset}) ds + e^{-\rho(T-t_k)} \phi(X_T^{k,x,\emptyset}) \right].$$

Algorithm 5.2 Backward construction of the approximate optimal strategy

1. Initialization at $k = K$.

Set

$$\hat{C}_K = \phi,$$

and define

$$\hat{M}_K(x) := \sup_{\xi \in D(x)} \left\{ c(x, \xi) + \hat{C}_K(x + \gamma(x, \xi)) \right\}.$$

Construct an approximate selector $\hat{\xi}_K$. In practice, both $\hat{M}_K$ and $\hat{\xi}_K$ are obtained by a uniform search over the compact set $D(x)$.

2. Backward induction.

For $k = K - 1, \dots, 0$ perform the following steps.

(a) Simulation step.

For each sample point $x_k^{(i)}$, run Algorithm 5.1 with inputs

$$x_k^{(i)}, \quad \{\hat{C}_j, \hat{M}_j, \hat{\xi}_j\}_{j=k+1}^K,$$

and generate one or several samples of $Y^{k, x_k^{(i)}}$.

Denote by $y_k^{(i)}$ the corresponding Monte Carlo estimate of

$$\mathbb{E}[Y^{k, x_k^{(i)}}].$$

(b) Regression step.

Fit the continuation operator $\hat{C}_k$ from the dataset

$$\{(x_k^{(i)}, y_k^{(i)})\}_{i=1}^{N_k}.$$

(c) Policy update.

Define

$$\hat{M}_k(x) := \sup_{\xi \in D(x)} \left\{ c(x, \xi) + \hat{C}_k(x + \gamma(x, \xi)) \right\}.$$

Construct an approximate maximizer $\hat{\xi}_k$ using a uniform search over a finite discretization of $D(x)$.

The resulting intervention rule satisfies

$$\hat{M}_k(x) \approx c(x, \hat{\xi}_k(x)) + \hat{C}_k(x + \gamma(x, \hat{\xi}_k(x))).$$

3. Output.

Return the family of continuation operators

$$(\hat{C}_0, \dots, \hat{C}_K)$$

Extension of Algorithms 5.1–5.2. Once the operators $(C_{k,m}, M_{k,m})$ are understood, the extension of the Algorithms 5.1 and 5.2 is entirely mechanical:

- Algorithm 5.1 now carries, along each trajectory, an additional variable m_j denoting the number of impulses still available at time t_j . At each grid point t_j , if an intervention is triggered, i.e. $M_{j,m_j}(X_{t_j}) > C_{j,m_j}(X_{t_j})$, the impulse cost is added, the state jumps, and the budget is updated to $m_{j+1} = m_j - 1$.

Otherwise, $m_{j+1} = m_j$. Except for this budget update, the simulation is identical to the unconstrained case.

- Algorithm 5.2 becomes two-dimensional: for each pair (k, m) , the targets for $C_{k,m}$ are obtained by rollouts of the simulation routine starting from (t_k, x) with m impulses remaining, and using $(C_{k+1,m}, M_{k+1,m}, C_{k,m-1}, M_{k,m-1})$ according to the budget update described above. Once $C_{k,m}$ is fitted, the corresponding intervention operator is

$$M_{k,m}(x) = \sup_{\xi \in D(x)} \{c(x, \xi) + C_{k,m-1}(x + \gamma(x, \xi))\}, \quad m \geq 1,$$

with $M_{k,0} \equiv +\infty$ to forbid interventions when no budget is left.

Both procedures preserve their structure. The only modification is that “continuation” refers to $(k+1, m)$ while “intervention” refers to $(k, m-1)$. Once this book-keeping is in place, the algorithms extend to the finite-impulse setting.

6. Numerical Example: The optimal dividend problem.. In this section, we illustrate the proposed methodology on an optimal dividend problem. An additional numerical example based on a harvesting problem is presented in Appendix C.

Classical models of optimal dividend distribution have been extensively studied in the literature; see, for instance, [41, 16, 22]. In the standard one-dimensional diffusion framework considered in [41, Example 6.4], the firm’s cash reserve evolves according to

$$dX_t = \mu dt + \sigma dW_t, \quad X_0 = x > 0.$$

At intervention times $(\tau_i)_{i \geq 1}$, the firm may distribute a dividend of size $\xi_i > 0$, incurring a fixed cost $c_f > 0$ and a proportional cost $\lambda \geq 0$. The post-intervention state is thus

$$X_{\tau_i} = X_{\tau_i^-} - (1 + \lambda)\xi_i - c_f, \quad 0 < (1 + \lambda)\xi_i + c_f \leq X_{\tau_i^-},$$

so that dividends are downward impulses which keep the reserve nonnegative. Each payment yields an immediate reward $e^{-\rho\tau_i}\xi_i$, where $\rho > 0$, and there is no running or terminal gain.

For a strategy $\alpha = (\tau_1, \tau_2, \dots; \xi_1, \xi_2, \dots)$, the controlled reserve process is

$$X_t^{(\alpha)} = x + \mu t + \sigma W_t - \sum_{\tau_i \leq t} \left((1 + \lambda)\xi_i + c_f \right), \quad t \geq 0,$$

and we define the bankruptcy time

$$\tau_S := \inf\{t > 0 : X_t^{(\alpha)} \leq 0\}.$$

The objective is to maximize the expected discounted dividend payments collected before bankruptcy:

$$J(\alpha)(x) := \mathbb{E}_x \left[\sum_{\tau_i < \tau_S} e^{-\rho\tau_i} \xi_i \right], \quad \Phi(x) := \sup_{\alpha} J(\alpha)(x).$$

Structure of the optimal policy.. As shown in [41, Example 6.4], the optimal dividend strategy exhibits the classical two-threshold pattern: there exist numbers $0 < \hat{x} < x^*$ such that

- if $X_t < x^*$, it is optimal to wait (*inaction region*);
- when X_t reaches x^* , it is optimal to pay a dividend that instantaneously brings the reserve down to $\hat{x}$.

The optimal dividend size at level $x \geq x^*$ is therefore

$$\zeta^*(x) = \frac{x - \hat{x} - c_f}{1 + \lambda}.$$

The value function is obtained by solving the second-order ordinary differential equation satisfied in the inaction region $(0, x^*)$, together with the appropriate boundary and smooth-fit conditions at the thresholds $\hat{x}$ and x^* . One finally verifies that the resulting candidate coincides with the value function by applying the verification theorem; see [41, Theorem 6.2]. We omit these standard computations, as they are classical and straightforward in this setting.

Extension to higher dimension.. A straightforward extension of this example to dimension d is obtained by considering d independent dividend problems, leading to a d -dimensional state process with decoupled dynamics and costs. From a numerical perspective, however, the problem becomes genuinely high-dimensional: the neural networks take as input a state vector in $\mathbb{R}^d$, and the optimization over intervention sizes is performed in dimension d . This provides a convenient testbed to assess the robustness and scalability of the algorithm in higher dimension.

We now examine the behaviour of our learned value functions and policies across several diagnostic experiments. All simulations use the same backward scheme described in Algorithm 5.2, either in the unconstrained or impulse-bounded regime. Note that since we work on a finite horizon, we impose bankruptcy at maturity

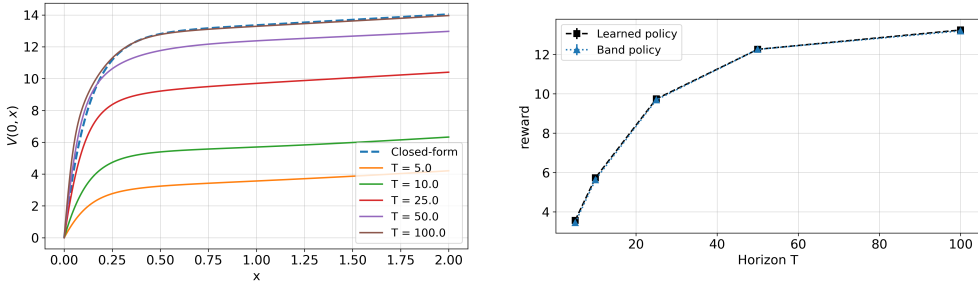


Fig. 1: Left ($d=1$): approximation of $V(0, x)$ for increasing maturities T . Right ($d=1$): comparison of the learned policy with the truncated band-policy.

Figure 1 highlights two key features. First, the approximation $V(0, x)$ converges to $\psi(x)$ as T increases, with an excellent match already for $T = 100$, while the backward iteration remains numerically stable. Second, the learned policy outperforms the truncated infinite-horizon band-policy for short maturities, reflecting the non-stationary nature of the finite-horizon problem. As T increases, both performances converge and become nearly indistinguishable, confirming the consistency of the learned strategy with the classical band-policy.

Figure 2 illustrates two complementary effects. First, $V^{(n)}(0, x)$ converges rapidly to $V(0, x)$, and for moderately large n the constrained and unconstrained solutions are nearly indistinguishable. Second, the corresponding controlled trajectories reflect this convergence: with few impulses the controller acts sparsely and paths remain higher,

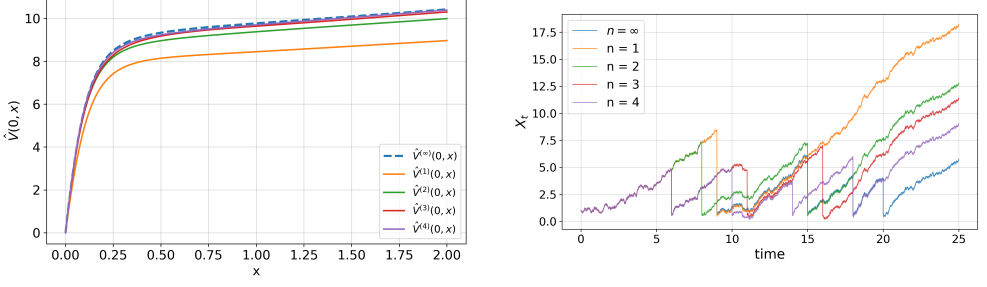


Fig. 2: Left (d=1): convergence of $V^{(n)}(0, x)$ to $V(0, x)$ as the number of impulses increases. Right (d=1): controlled trajectories for different impulse budgets.

whereas increasing the impulse budget leads to more frequent interventions and trajectories that stay closer to the optimal band, eventually matching the unconstrained behaviour.

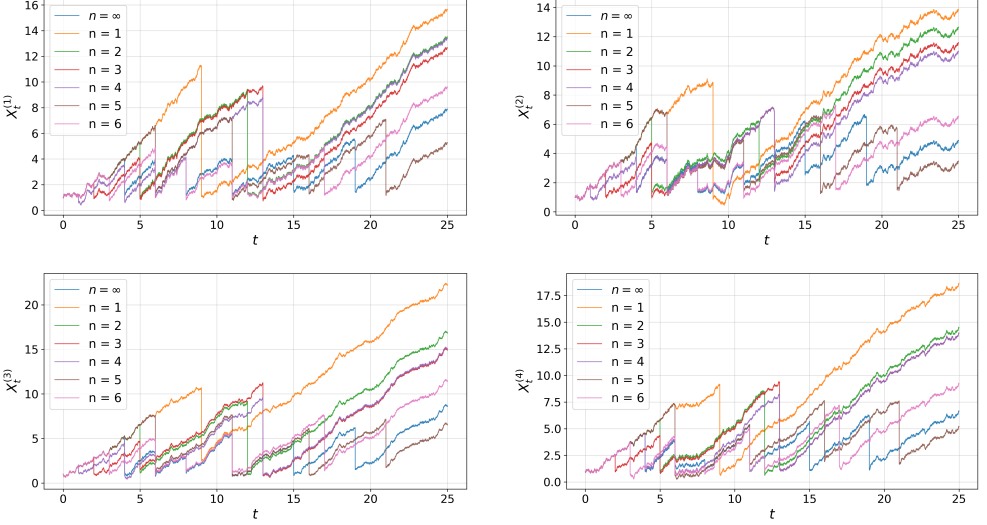


Fig. 3: Controlled trajectories for the four coordinates of the 4-dimensional system under different impulse budgets.

Figure 3 displays the controlled trajectories for each coordinate of the system in dimension 4. As the number of allowed impulses increases, a band-type policy clearly emerges on each coordinate: the process is repeatedly pushed back toward a lower threshold after reaching an upper level. A key difference with the one-dimensional case is that the impulse constraint is global rather than coordinate-wise. As a result, interventions tend to act simultaneously on all coordinates, even when some components do not require adjustment. This coupling effect explains why certain trajectories are occasionally over-corrected.

Figure 4 reports the same diagnostics as in the one-dimensional case, now in dimension $d = 6$. While the qualitative behaviour remains similar, a notable dif-

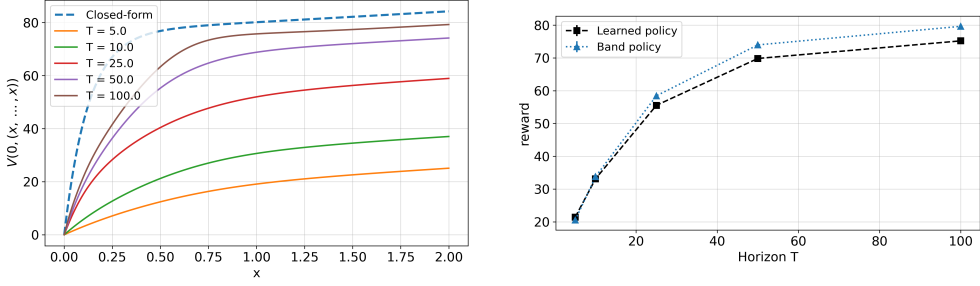


Fig. 4: Left ($d=6$): approximation of $V(0, (x, \dots, x))$ for increasing maturities T . Right ($d=6$): comparison of the learned policy with the truncated band-policy.

ference appears: the approximation no longer converges exactly to the closed-form solution as T increases. This discrepancy reflects the increased numerical difficulty of the high-dimensional setting, in particular in the joint optimization over impulses. Nevertheless, the learned policy retains the correct qualitative structure and remains close to the band-policy benchmark.

Remark 6.1. A second numerical example devoted to the well known harvesting problem [4, 37] is reported in Appendix C.

7. Conclusion. In this work, we developed a regression-based numerical framework for stochastic impulse control problems with controls taking values in a compact subset of $\mathbb{R}^q$. The construction combines probabilistic dynamic programming techniques with neural-network-based approximations and randomized optimization procedures. From a theoretical perspective, we first established the well-posedness of the controlled dynamics under general Lipschitz assumptions, proving existence and uniqueness of the associated controlled process. We then showed that the original impulse control problem can be approximated by restricting the admissible set to strategies with finitely many interventions. To address the discretization of intervention times, we introduced the class of δ -separated strategies, excluding clustering phenomena and allowing the derivation of short-time stability estimates. We proved that every δ -separated strategy can be approximated by a grid-valued strategy whenever the mesh size satisfies $|\pi| \leq \delta$, with an approximation error of order $O(|\pi|^{1/2})$.

Once the problem is reduced to a finite intervention grid, we derived a discrete-time dynamic programming formulation of the impulse control problem. This led to an explicit construction of canonical optimal strategies in terms of stopping rules and intervention regions. More precisely, the optimal controls were characterized through recursively defined hitting times associated with the intervention regions, together with a verification lemma. Building on this representation, we introduced a backward numerical algorithm inspired by regression Monte Carlo methods in the spirit of [47, 38], and closely related to the framework developed by [39]. The continuation values are approximated recursively by neural-network regressors, while the intervention operator is approximated through randomized uniform search over the control space. We also extended the methodology to the case where the controller is constrained to perform only finitely many interventions.

Relying on the results of [34], we further established convergence estimates for the randomized approximation of the intervention operator when continuation values are

615 represented by piecewise affine neural networks. This provides a theoretical control
616 of the optimization error induced by the randomization procedure.

617 Finally, the numerical experiments illustrate the robustness and flexibility of the
618 proposed methodology. The one-dimensional benchmark problems confirm the accu-
619 racy and stability of the approach, while the higher-dimensional experiments demon-
620 strate that the framework remains effective beyond the low-dimensional setting, with
621 satisfactory results obtained in dimensions up to six.

Appendix A. Technical Proofs.

A.1. Proof of Proposition 2.1.

Proof. Fix $\alpha = (\tau^i, \xi^i)_{i \geq 1} \in \mathcal{A}$ and, for $n \geq 1$, define the truncated strategies

$$(\tau_n^i, \xi_n^i) = \begin{cases} (\tau^i, \xi^i), & i \leq n, \\ (\tau^i \vee (T+1), \xi^i), & i > n. \end{cases}$$

Then $\alpha^n := (\tau_n^i, \xi_n^i)_{i \geq 1} \in \mathcal{A}$ and there are only finitely many impulses on $[0, T]$. By Assumption 1 (global Lipschitz of b, σ and boundedness of γ), the SDE with finitely many jumps admits a unique strong solution $X^{\alpha^n} \in \mathbf{S}_{\mathcal{F}}^2[0, T]$.

Fix $n, p \geq 1$. Using 2.2 for X^{α^n} and $X^{\alpha^{n+p}}$, we write for $t \in [0, T]$:

$$\begin{aligned} X_t^{\alpha^n} - X_t^{\alpha^{n+p}} &= \int_0^t \left(b(s, X_s^{\alpha^n}) - b(s, X_s^{\alpha^{n+p}}) \right) ds \\ &\quad + \int_0^t \left(\sigma(s, X_s^{\alpha^n}) - \sigma(s, X_s^{\alpha^{n+p}}) \right) dW_s + J_t. \end{aligned}$$

where

$$J_t := \sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq t\}} \gamma \left(X_{\tau^i-}^{\alpha^{n+p}}, \xi^i \right),$$

and by the boundedness of γ ,

$$|J_t| \leq \bar{\gamma} \sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq t\}}.$$

By the Lipschitz continuity of b and σ , together with the Burkholder–Davis–Gundy inequality, there exists a constant $\Lambda \geq 0$ depending only on L, T , and the BDG constant such that

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |X_s^{\alpha^n} - X_s^{\alpha^{n+p}}|^2 \right] \leq \Lambda \int_0^t \mathbb{E} \left[\sup_{0 \leq u \leq s} |X_u^{\alpha^n} - X_u^{\alpha^{n+p}}|^2 \right] ds + 3 \mathbb{E} \left[\sup_{0 \leq s \leq t} |J_s|^2 \right].$$

Using the bound on J and the fact that $t \mapsto \sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq t\}}$ is nondecreasing, we obtain

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |J_s|^2 \right] \leq \bar{\gamma}^2 \mathbb{E} \left[\left(\sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq T\}} \right)^2 \right].$$

Hence,

$$\begin{aligned} \mathbb{E} \left[\sup_{0 \leq s \leq t} |X_s^{\alpha^n} - X_s^{\alpha^{n+p}}|^2 \right] &\leq \Lambda \int_0^t \mathbb{E} \left[\sup_{0 \leq u \leq s} |X_u^{\alpha^n} - X_u^{\alpha^{n+p}}|^2 \right] ds \\ &\quad + 3\bar{\gamma}^2 \mathbb{E} \left[\left(\sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq T\}} \right)^2 \right]. \end{aligned}$$

By Gronwall's lemma, this yields for $t = T$,

$$\mathbb{E} \left[\sup_{0 \leq s \leq T} |X_s^{\alpha^n} - X_s^{\alpha^{n+p}}|^2 \right] \leq 3\bar{\gamma}^2 e^{\Lambda T} \mathbb{E} \left[\left(\sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq T\}} \right)^2 \right],$$

647 that is,

$$648 \quad \mathbb{E} \left[\sup_{0 \leq s \leq T} |X_s^{\alpha^n} - X_s^{\alpha^{n+p}}|^2 \right] \leq C \mathbb{E} \left[\left(\sum_{i=n+1}^{n+p} \mathbf{1}_{\{\tau^i \leq T\}} \right)^2 \right], \quad C := 3\bar{\gamma}^2 e^{\Lambda T}.$$

649 Since $\alpha \in \mathcal{A}$, the right-hand side tends to 0 as $n \rightarrow \infty$ uniformly in p . Thus $(X^{\alpha^n})_{n \geq 1}$
 650 is Cauchy in $\mathbf{S}_{\mathcal{F}}^2[0, T]$ and converges to some $\bar{X}$. By construction of the truncations,
 651 $\bar{X} = X^{\alpha^n}$ on $[0, \tau^n]$, and $\tau^n \rightarrow +\infty$ a.s., so $\bar{X}$ satisfies 2.2. Uniqueness is obtained
 652 analogously. $\square$

653 A.2. Proof of Proposition 2.2.

654 *Proof.* The sequence $(V_0^n)_{n \in \mathbb{N}}$ is non-decreasing and bounded above by V_0 , hence

$$655 \quad V_0^\infty := \lim_{n \rightarrow \infty} V_0^n \leq V_0.$$

656 We prove the reverse inequality. Fix $\varepsilon > 0$ and choose $\alpha = (\tau^i, \xi^i) \in \mathcal{A}$ such that

$$657 \quad (\text{A.1}) \quad J(\alpha) \geq V_0 - \varepsilon.$$

658 For each $n \geq 1$, define the truncated strategy $\alpha^n = (\tau_n^i, \xi_n^i)$ by

$$659 \quad (\tau_n^i, \xi_n^i) = \begin{cases} (\tau^i, \xi^i), & i \leq n, \\ (\tau^i \vee (T+1), \xi^i), & i > n. \end{cases}$$

660 We decompose

$$661 \quad |J(\alpha) - J(\alpha^n)| \leq \mathbb{E} \left| g(X_T^\alpha) - g(X_T^{\alpha^n}) \right| + \mathbb{E} \int_0^T \left| f(t, X_t^\alpha) - f(t, X_t^{\alpha^n}) \right| dt$$

$$662 \quad + \mathbb{E} \left| \sum_{0 \leq \tau^i \leq T} c(X_{\tau^i-}^\alpha, \xi^i) - \sum_{0 \leq \tau_n^i \leq T} c(X_{\tau_n^i-}^{\alpha^n}, \xi_n^i) \right|.$$

663 By Lipschitz continuity,

$$664 \quad \mathbb{E} \left| g(X_T^\alpha) - g(X_T^{\alpha^n}) \right| \leq L \|X^\alpha - X^{\alpha^n}\|_{\mathbf{S}_{\mathcal{F}}^2},$$

665

$$666 \quad \mathbb{E} \int_0^T \left| f(t, X_t^\alpha) - f(t, X_t^{\alpha^n}) \right| dt \leq LT \|X^\alpha - X^{\alpha^n}\|_{\mathbf{S}_{\mathcal{F}}^2}.$$

667 For the impulse part, since α^n coincides with α up to the n -th intervention, the
 668 difference involves only the tail $i \geq n+1$:

$$669 \quad \left| \sum_{0 \leq \tau^i \leq T} c(X_{\tau^i-}^\alpha, \xi^i) - \sum_{0 \leq \tau_n^i \leq T} c(X_{\tau_n^i-}^{\alpha^n}, \xi_n^i) \right| = \sum_{i \geq n+1} \mathbf{1}_{\{\tau^i \leq T\}} |c(X_{\tau^i-}^\alpha, \xi^i)|.$$

Using Assumption 2 (iii) and $|c(x, z)| \leq |c(0, z)| + L|x|$, we get

$$\begin{aligned} \mathbb{E} \left[\sum_{i \geq n+1} \mathbf{1}_{\{\tau^i \leq T\}} |c(X_{\tau^i-}^\alpha, \xi^i)| \right] &\leq \sup_{z \in A} |c(0, z)| \mathbb{E} \left[\sum_{i \geq n+1} \mathbf{1}_{\{\tau^i \leq T\}} \right] \\ &\quad + L \mathbb{E} \left[\sum_{i \geq n+1} \mathbf{1}_{\{\tau^i \leq T\}} |X_{\tau^i-}^\alpha| \right] \\ &\leq \sup_{z \in A} |c(0, z)| \mathbb{E}[N_T^{(n)}] \\ &\quad + L \mathbb{E} \left[N_T^{(n)} \sup_{0 \leq s \leq T} |X_s^\alpha| \right]. \end{aligned}$$

where $N_T^{(n)} := \sum_{i \geq n+1} \mathbf{1}_{\{\tau^i \leq T\}}$. By Cauchy–Schwarz inequality,

$$\mathbb{E}[N_T^{(n)}] \leq (\mathbb{E}[(N_T^{(n)})^2])^{1/2}, \quad \mathbb{E}[N_T^{(n)} \sup_{s \leq T} |X_s^\alpha|] \leq (\mathbb{E}[(N_T^{(n)})^2])^{1/2} \|X^\alpha\|_{\mathbf{S}^2}.$$

Collecting all bounds, there exists a constant $\Lambda > 0$ (depending only on L, T and $\sup_{z \in A} |c(0, z)|$) such that

$$|J(\alpha) - J(\alpha^n)| \leq \Lambda \|X^\alpha - X^{\alpha^n}\|_{\mathbf{S}^2} + \Lambda (1 + \|X^\alpha\|_{\mathbf{S}^2}) \left(\mathbb{E}[(N_T^{(n)})^2] \right)^{1/2}.$$

Since $\alpha \in \mathcal{A}$, we have $N_T := \sum_{i \geq 1} \mathbf{1}_{\{\tau^i \leq T\}} \in L^2$. Hence

$$0 \leq N_T^{(n)} := \sum_{i \geq n+1} \mathbf{1}_{\{\tau^i \leq T\}} \downarrow 0 \quad \text{a.s.}$$

and $(N_T^{(n)})^2 \leq N_T^2 \in L^1$, so by dominated convergence

$$\mathbb{E}[(N_T^{(n)})^2] \xrightarrow{n \rightarrow \infty} 0.$$

Moreover, by Proposition 2.1, $\|X^{\alpha^n} - X^\alpha\|_{\mathbf{S}^2} \rightarrow 0$. Plugging these limits into the previous estimate yields

$$|J(\alpha) - J(\alpha^n)| \xrightarrow{n \rightarrow \infty} 0.$$

This convergence with (A.1) gives $V_0^\infty \geq V_0 - \epsilon$. Since $\epsilon > 0$ is arbitrarily chosen we get $V_0^\infty \geq V_0$. $\square$

A.3. Proof of Lemma 3.4.

Proof. Fix $n \in \mathbb{N}$ and $\alpha \in \mathcal{A}_n$. By definition of $\mathcal{A}_n$,

$$N_T(\alpha) := \sum_{i \geq 1} \mathbf{1}_{\{\tau^i \leq T\}} \leq n$$

Recall that X^α solves

$$X_t^\alpha = X_0 + \int_0^t b(s, X_s^\alpha) ds + \int_0^t \sigma(s, X_s^\alpha) dW_s + \sum_{0 \leq \tau^i \leq t} \gamma(X_{\tau^i-}^\alpha, \xi^i), \quad 0 \leq t \leq T.$$

691 Set

$$692 \quad A_t := \int_0^t b(s, X_s^\alpha) ds, \quad M_t := \int_0^t \sigma(s, X_s^\alpha) dW_s, \quad J_t := \sum_{0 \leq \tau^i \leq t} \gamma(X_{\tau^i-}^\alpha, \xi^i),$$

693 so that $X_t^\alpha = X_0 + A_t + M_t + J_t$ and

$$694 \quad \sup_{0 \leq t \leq T} |X_t^\alpha|^2 \leq 4 \left(|X_0|^2 + \sup_{0 \leq t \leq T} |A_t|^2 + \sup_{0 \leq t \leq T} |M_t|^2 + \sup_{0 \leq t \leq T} |J_t|^2 \right).$$

695 By Cauchy–Schwarz and linear growth of b , there exists a constant $\Lambda_1 \geq 0$ such that

$$696 \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} |A_t|^2 \right] \leq \Lambda_1 \int_0^T \left(1 + \mathbb{E} \left[\sup_{0 \leq u \leq s} |X_u^\alpha|^2 \right] \right) ds.$$

697 By the Burkholder–Davis–Gundy inequality and linear growth of σ , there exists $\Lambda_2 \geq$
698 0 such that

$$699 \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} |M_t|^2 \right] \leq \Lambda_2 \int_0^T \left(1 + \mathbb{E} \left[\sup_{0 \leq u \leq s} |X_u^\alpha|^2 \right] \right) ds.$$

700 Finally, by boundedness of γ and since $N_T(\alpha) \leq n$,

$$701 \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} |J_t|^2 \right] \leq \bar{\gamma}^2 n^2.$$

702 Combining these estimates and setting $\Lambda_3 := 4(\Lambda_1 + \Lambda_2)$, we obtain

$$703 \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^\alpha|^2 \right] \leq 4 \mathbb{E}[|X_0|^2] + \Lambda_3 \int_0^T \left(1 + \mathbb{E} \left[\sup_{0 \leq u \leq s} |X_u^\alpha|^2 \right] \right) ds + 4\bar{\gamma}^2 n^2.$$

704 Define

$$705 \quad \Phi(t) := \mathbb{E} \left[\sup_{0 \leq u \leq t} |X_u^\alpha|^2 \right], \quad t \in [0, T].$$

706 Then

$$707 \quad \Phi(T) \leq \Lambda_4 + \Lambda_3 \int_0^T \Phi(s) ds, \quad \Lambda_4 := 4 \mathbb{E}[|X_0|^2] + \Lambda_3 T + 4\bar{\gamma}^2 n^2.$$

708 By Gronwall’s lemma,

$$709 \quad \Phi(T) \leq \Lambda_4 e^{\Lambda_3 T}.$$

710 Setting

$$711 \quad \Lambda := \Lambda_4 e^{\Lambda_3 T} = e^{\Lambda_3 T} \left(4 \mathbb{E}[|X_0|^2] + \Lambda T + 4\bar{\gamma}^2 n^2 \right),$$

712 we obtain

$$713 \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^\alpha|^2 \right] \leq \Lambda,$$

714 with Λ depending only on $L, K, T, \bar{\gamma}, n$ and $\mathbb{E}[|X_0|^2]$. Since the argument is valid for
715 any $\alpha \in \mathcal{A}_n$, the claim follows. $\square$

716 **Appendix B. Convergence of maximization through randomization.**

717 The numerical evaluation of the optimum (4.1) introduces an additional difficulty,
718 as it requires optimizing over the control variable on top of the multiple stopping

structure already present. Traditional maximization methods in reinforcement learning, like actor-critic and policy-gradient methods (e.g. DDPG [36] or PPO [44]) as well as derivative-free evolutionary algorithms like CMA-ES [25], can be employed for this purpose. However, these methods are considerably more complex than the randomization-based strategy considered in this work. Following the approach introduced in [34], we approximate $y^* = \sup_{z \in A} G(z)$ by evaluating G at i.i.d. uniform samples $\xi_1, \dots, \xi_M \sim \text{Unif}(A)$ and using the sample maximum

$$Y_{(M)} := \max_{1 \leq i \leq M} G(\xi_i).$$

B.1. Asymptotic and non-asymptotic convergence result. Let F denote the distribution of $Y_i = G(\xi_i)$. Extreme Value Theory provides the appropriate framework for characterizing the asymptotic behavior of the sample maximum. We seek constants $a_M > 0$ such that

$$a_M^{-1}(Y_{(M)} - y^*) \rightarrow H, \quad m \rightarrow \infty,$$

for some non-degenerate limit distribution H . For neural networks with Leaky-ReLU activation

$$(B.1) \quad \text{activation}(z) = s_+ \max\{z, 0\} + s_- \min\{z, 0\}, \quad z \in \mathbb{R},$$

where $s_+ > 0$ and $s_- > 0$ and uniform input distribution, the authors of [1] show that the distribution function F is continuous and piecewise polynomial, with each piece of degree at most q . Moreover, if y^* is a root of multiplicity θ of the polynomial $1 - F(y)$ near the upper boundary, then

$$1 - F(y) = O((y^* - y)^\theta), \quad y \uparrow y^*.$$

This polynomial tail behavior places F in the maximum domain of attraction of the Weibull distribution with $a_M = O(M^{-1/\theta})$.

In [34], the authors also gave a non-asymptotic error estimate based on the mean-square error

$$\mathbb{E}[R_M^2], \quad R_M := Y_{(M)} - y^*.$$

Under the polynomial tail assumption of [1], one obtains the finite-sample bound

$$(B.2) \quad \mathbb{E}[R_M^2] \leq (y^* - y_*)^2 (K_1)^M + K_2 M \text{Beta}(1 + 2/\theta, M),$$

where K_1 and K_2 are two positive constants with $K_1 < 1$, $y_* = \inf_{z \in A} G(z)$, and Beta denotes the Beta function defined by

$$(B.3) \quad \text{Beta}(p_1, p_2) := \int_0^1 x^{p_1-1} (1-x)^{p_2-1} dx.$$

This yields the non-asymptotic rate

$$\mathbb{E}[R_M^2] = O(M^{-2/\theta}).$$

Thus, the convergence is faster when $1 - F$ has a lower-order zero at y^* .

Sharper rates arise when the density f of F is sufficiently regular near y^* . If f exists, is continuously differentiable in a neighborhood of y^* , and satisfies $f(y^*) > 0$,

the classical results of [23] give the non-asymptotic estimate (cf. [34])

$$\mathbb{E}[R_M^2] \leq \frac{4}{f(y^*)^2(M+1)(M+2)} + \left((y^* - y_*)^2 + \frac{1}{f^2(y^*)} \right) M e^{-M^{1-\delta}} \\ + \frac{M^{-4\delta}}{4} \left(\sup_{y \in Q_M} \frac{(f'(y))^2}{f^6(y)} \right).$$

where $Q_M = (q_{1-M^{-\delta}}, y^*]$ with $q_{1-M^{-\delta}}$ is the quantile of level $1 - M^{-\delta}$ and $0 < \delta < 1$, which implies the accelerated rate

$$\mathbb{E}[R_M^2] = O(M^{-2}).$$

This improves upon earlier quantile-approximation bounds such as [7] and demonstrates that randomized maximization can achieve high precision.

B.2. Application to the computation of intervention marks. Randomized evaluation provides an efficient and theoretically grounded approach for maximizing neural network surrogates. Under mild conditions, the sample maximum $Y_{(M)}$ converges to the true maximum y^* at a rate determined by the local polynomial structure of F near y^* .

Fix a pair $(t_k, i) \in (\alpha \cup \{T+1\}) \times \{0, \dots, n-1\}$, where the index i records that the controller is allowed to execute the i -th impulse. Let v_k^i denote the corresponding value function at time t_k under this intervention capacity, and let $G(\cdot)$ be the Leaky-ReLU neural network used to approximate v_k^i on the domain D_0 . For each $x \in D_0$, define

$$(B.4) \quad y^*(x) := \max_{\xi \in A} \{G(x + \gamma(x, \xi)) + c(x, \xi)\}; \\ y_*(x) := \min_{\xi \in A} \{G(x + \gamma(x, \xi)) + c(x, \xi)\}.$$

For $\xi_1, \dots, \xi_M$ i.i.d. uniformly distributed over the set A , let $Y_{(M)}(x)$ denote the sample maximum of $G(x + \gamma(x, \xi_1)) + c(x, \xi_1), \dots, G(x + \gamma(x, \xi_M)) + c(x, \xi_M)$; let $F(y; x)$ and $f(y; x)$ be respectively the CDF and the left-continuous version of the density function of $G(x + \gamma(x, \xi_1)) + c(x, \xi_1)$.

Assumption 3. The set A is a polytope. For each $x \in D_0$, the mappings

$$\gamma(x, \cdot) \quad \text{and} \quad c(x, \cdot)$$

are continuous and piecewise affine on A , that is, there exists a finite polyhedral partition of A such that both functions are affine on each element of the partition.

LEMMA B.1. (i) Under Assumption 3, there exist positive constants $K_1(x) \in (0, 1)$ and $K_2(x)$, as well as a positive integer $\theta(x) \in \{1, \dots, q\}$, such that

$$(B.5) \quad \mathbb{E} \left[(y^*(x) - Y_{(M)}(x))^2 \right] \leq (y^*(x) - y_*(x))^2 K_1(x)^M \\ + K_2(x) M \text{Beta} \left(1 + \frac{2}{\theta(x)}, M \right).$$

for every positive integer M .

(ii) Let us assume that the control space dimension $q = 1$ and that Assumption 3 is

788 true. Let $\epsilon(x) \in (0, y^*(x) - y_*(x)]$ be the breaking point of the piecewise affine CDF
 789 $F(y; x)$ such that $F(y; x)$ is affine on $[y^*(x) - \epsilon(x), y^*(x)]$. Then for any positive real
 790 number $\delta \in (0, 1)$ and for every positive integer $M > (1 - F(y^*(x) - \epsilon(x); x))^{-1/\delta}$, it
 791 holds that

$$(B.6) \quad \mathbb{E} \left[(y^*(x) - Y_{(M)}(x))^2 \right] \leq \frac{4}{f^2(y^*(x); x)(M+1)(M+2)} + \left((y^*(x) - y_*(x))^2 + \frac{1}{f^2(y^*(x); x)} \right) M e^{-M^{1-\delta}}.$$

793 *Proof.* Since both $G(\cdot)$ and $\gamma(x, \cdot)$ are piecewise affine, their composition $G(x +$
 794 $\gamma(x, \cdot))$ is also piecewise affine. By Theorem 2.1 of [1], the CDF of $G(x + \gamma(x, \xi_1)) +$
 795 $c(x, \xi_1)$ is therefore piecewise polynomial, with polynomial degree at most the dimension
 796 q of the set A . Let $\theta(x) \leq q$ denote the multiplicity of the root of this CDF
 797 at $y^*(x)$. With this choice of $\theta(x)$, Assumption 3.3 (ii) in [34] is satisfied. The estimate
 798 (B.5) follows directly from Theorem 3.5 of [34]. The estimate (B.6) follows from
 799 Theorem 3.8 of [35] and the choice of ϵ_F in its proof. $\square$

800 *Remark B.2.* In many impulse control problems, $\gamma(x, \cdot)$ and $c(x, \cdot)$ are piecewise
 801 affine. Otherwise, they may be approximated by neural networks with Leaky-ReLU
 802 activations.

803 *Assumption 4.* (i) $\sup_{x \in D_0} y^*(x) =: y^* < +\infty$, $\inf_{x \in D_0} y_*(x) =: y_* >$
 804 $-\infty$.
 805 (ii) $\sup_{x \in D_0} K_1(x) =: K_1 < 1$, $\sup_{x \in D_0} K_2(x) =: K_2 < +\infty$.
 806 (iii) $\inf_{x \in D_0} f(y_*(x); x) =: \underline{f} > 0$, $\sup_{x \in D_0} F(y^*(x) - \epsilon(x); x) =: \bar{F} < 1$.

807 *Remark B.3.* (i) If x is in a bounded domain D_0 , Assumption 4 (i) is true. **why?**
 808 (ii) Assuming the domain D_0 bounded, to verify Assumption 4 (ii), it suffices to verify
 809 the **continuity** of $K_1(x)$ and $K_2(x)$ in x , along the lines of [1] and [34].
 810 (iii) Under Assumption 3, if each affine expression of $\gamma(x, \cdot)$ and $c(x, \cdot)$ has coefficients
 811 continuous in x , then according to procedures of computing the CDF of the neural
 812 network output in [1] and the vertices in [2], the coefficients of the CDF and the
 813 coordinates of the vertices are continuous in x . The breaking points of the piecewise
 814 polynomial CDF, which are the “levels” defined in Definition A.6 of [1], are continuous
 815 in x . When $q = 1$, the density function, which is the slope of the CDF, is also
 816 continuous in x . If in addition the domain D_0 is bounded, Assumption 4 (iii) is true.

817 **THEOREM B.4.** (i) Under the Assumptions 3-4 (i)(ii), with X_t^α taking its values
 818 in D_0 ,

$$(B.7) \quad \mathbb{E} \left[(y^*(X_t^\alpha) - Y_{(M)}(X_t^\alpha))^2 \right] \leq O(M^{-2/q}).$$

820 (ii) Under the Assumptions 3-4(i)(iii), and assuming that $q = 1$, with X_t^α taking its
 821 values in D_0 ,

$$(B.8) \quad \mathbb{E} \left[(y^*(X_t^\alpha) - Y_{(M)}(X_t^\alpha))^2 \right] \leq O(M^{-2}).$$

823 *Proof.* (i) By Lemma B.1 (i) and Assumptions 4 (i)(ii), $\mathbb{E} \left[(y^*(x) - Y_{(M)}(x))^2 \right]$
 824 has a uniform upper bound in x , expressed as

$$(B.9) \quad \sup_{x \in D_0} \mathbb{E} \left[(y^*(x) - Y_{(M)}(x))^2 \right] \leq (y^* - y_*)^2 K_1^M + K_2 M \text{Beta}(1 + 2/q, M).$$

826 When $M \rightarrow \infty$, $K_1^M = o(M^{-2/q})$ since $K_1 \in (0, 1)$, and $Beta(1 + 2/q, M) =$
 827 $O(M^{-(1+2/q)})$. The estimation (B.7) follows from conditioning on X_t^α .
 828 (ii) By Lemma B.1 (ii) and Assumption 4 (i)(iii), for every positive integer $M >$
 829 $(1 - \bar{F})^{-1/\delta}$, $\mathbb{E} \left[(y^*(x) - Y_{(M)}(x))^2 \right]$ has a uniform upper bound in x , expressed as

$$\begin{aligned} \sup_{x \in D_0} \mathbb{E} \left[(y^*(x) - Y_{(M)}(x))^2 \right] &\leq \frac{4}{\underline{f}^2(M+1)(M+2)} \\ &+ \left((y^* - y_*)^2 + \frac{1}{\underline{f}^2} \right) M e^{-M^{1-\delta}}. \end{aligned} \quad (\text{B.10})$$

831 For every $\delta \in (0, 1)$, $M e^{-M^{1-\delta}} = o(M^{-2})$. The estimation (B.8) also follows from
 832 conditioning on X_t^α . $\square$

833 *Remark B.5.* Although the approximation accuracy of the neural network for the
 834 value function v would typically be expected to depend on the dimension m of the
 835 underlying process X , it should be noted that because of (B.7) the accuracy of the
 836 randomized maximum is, in fact, independent of m . Instead, it depends solely on the
 837 dimension q of the control space. Consequently, provided that the high-dimensional
 838 process X is governed by a low-dimensional control variable ξ , say $q \leq 3$, the ran-
 839 domized maximum exhibits sufficiently fast convergence.

840 **Appendix C. Numerical Example: The harvesting problem.**

841 Optimal harvesting under stochastic fluctuations has been widely investigated in
 842 the literature; see, for example, [4, 5], and the recent one-sided impulse formulations
 843 in [37]. We consider here a simple one-dimensional diffusion with downward impulses
 844 in the spirit of [41, Example 6.5].

845 We consider the uncontrolled diffusion

$$846 \quad dX_t = \mu dt + \sigma dW_t, \quad X_0 = x \in \mathbb{R},$$

847 and one-sided impulse controls $\alpha = (\tau_1, \tau_2, \dots; \xi_1, \xi_2, \dots)$ with downward impulses
 848 $\xi_k > 0$, acting through

$$849 \quad X^{(\alpha)}(t) = x + \mu t + \sigma W_t - \sum_{\tau_k \leq t} \xi_k.$$

850 The state 0 is absorbing, and only downward impulses are admissible. The objective
 851 is to *minimize* over the strategies α the discounted running and intervention costs:

$$852 \quad J(\alpha)(s, x) = \mathbb{E}_x \left[\int_0^\infty e^{-\rho(s+t)} \zeta (X_t^{(\alpha)} - x_0)^2 dt + \sum_k e^{-\rho(s+\tau_k)} (c + \lambda \xi_k) \right].$$

$$853 \quad \Phi(s, x) := \min_{\alpha} J(\alpha)(s, x),$$

855 with $\zeta \geq 0$ and $x_0 \in \mathbb{R}$.

856 *Structure of the optimal policy.* As in the dividend problem 6, the value function
 857 is obtained by solving a second-order ordinary differential equation, together with
 858 appropriate boundary and smooth-fit conditions. The optimality of the candidate
 859 solution is then verified using [41, Theorem 6.2]. The resulting optimal strategy
 860 again exhibits a band-type structure, with impulses used to keep the process within
 861 a suitable region.

As in the dividend problem 6, this setting can be naturally extended to dimension d by considering independent components. The numerical experiments are performed using Algorithm 5.2.

Figure 5 exhibits the same qualitative behaviour as in the dividend setting. The value function stabilizes as T increases, and the learned policy aligns with the band-policy benchmark, with only minor discrepancies.

Figure 6 also confirms that the bounded-impulse approximation is rapidly accurate: as the number of allowed interventions increases, both the value function and the controlled trajectories converge toward their unconstrained counterparts, with the policy progressively enforcing the band structure.

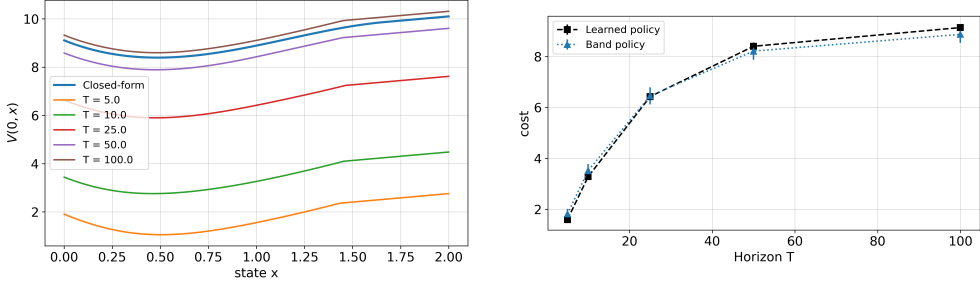


Fig. 5: Left ($d=1$): approximation of $V(0, x)$ for increasing maturities T . Right ($d=1$): comparison of the learned policy with the truncated band-policy.

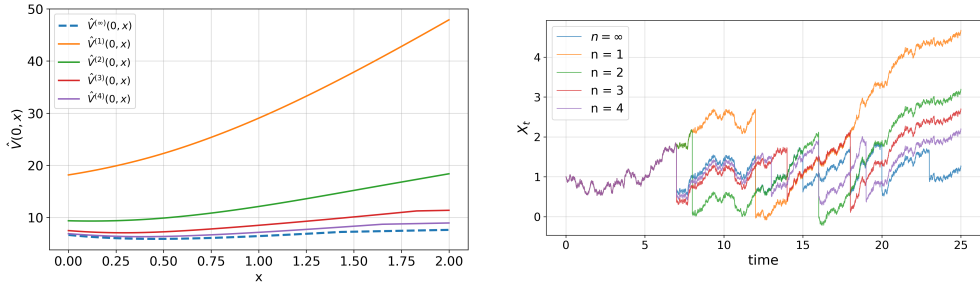


Fig. 6: Left ($d=1$): convergence of $V^{(n)}(0, x)$ to $V(0, x)$ as the number of impulses increases. Right ($d=1$): controlled trajectories for different impulse budgets.

Figure 7 shows the controlled trajectories for each coordinate in dimension $d = 4$. As the impulse budget increases, a band-type behaviour emerges on each component, with the process repeatedly pushed back toward a target region. As in the dividend case, the global constraint on the number of interventions induces simultaneous corrections across coordinates, even when some components are already close to their optimal level.

Figure 8 confirms that the qualitative behaviour observed in dimension one carries over to higher dimension. The value function stabilizes as T increases and the learned policy remains close to the band-policy benchmark, although convergence becomes less accurate due to the increased difficulty of the high-dimensional optimization.

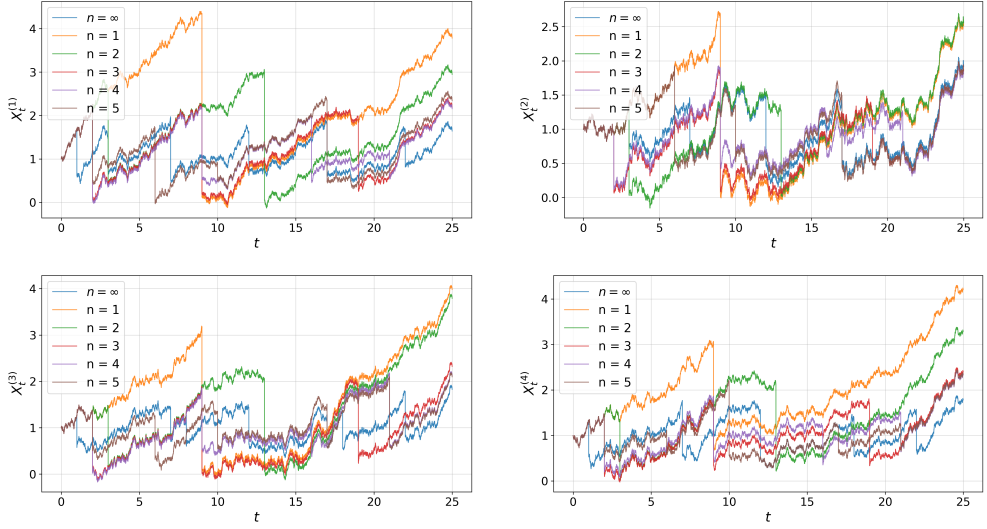


Fig. 7: Controlled trajectories for the four coordinates of the 4-dimensional system under different impulse budgets.

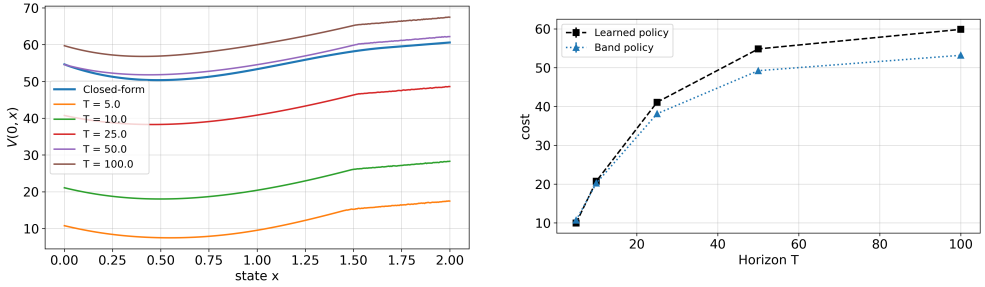


Fig. 8: Left ($d=4$): approximation of $V(0, (x, \dots, x))$ for increasing maturities T . Right ($d=6$): comparison of the learned policy with the truncated band-policy.

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